Measurement I

From concepts to numbers

Sebastian Ramirez-Ruiz
European University Institute (EUI)
October 23, 2026

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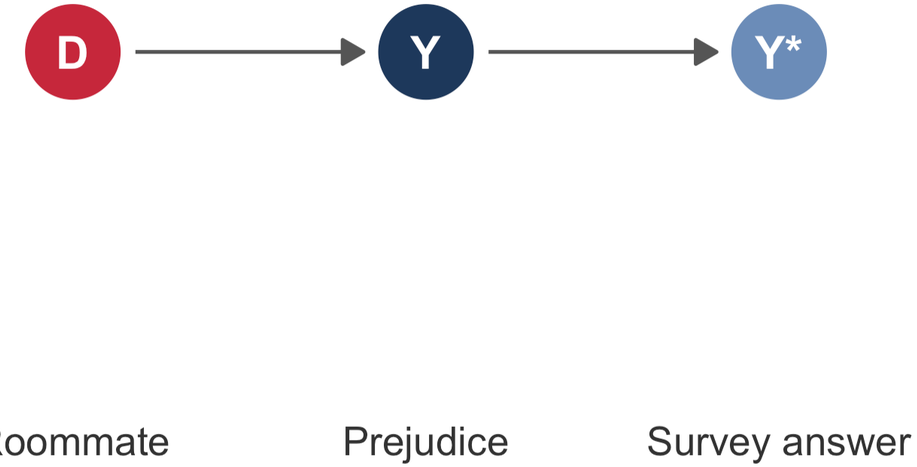
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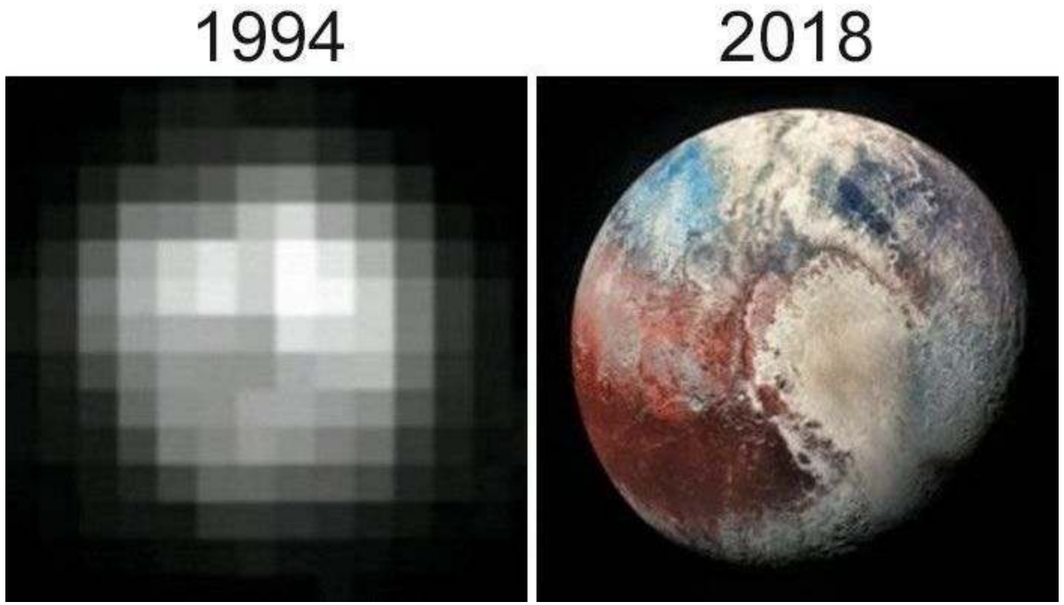
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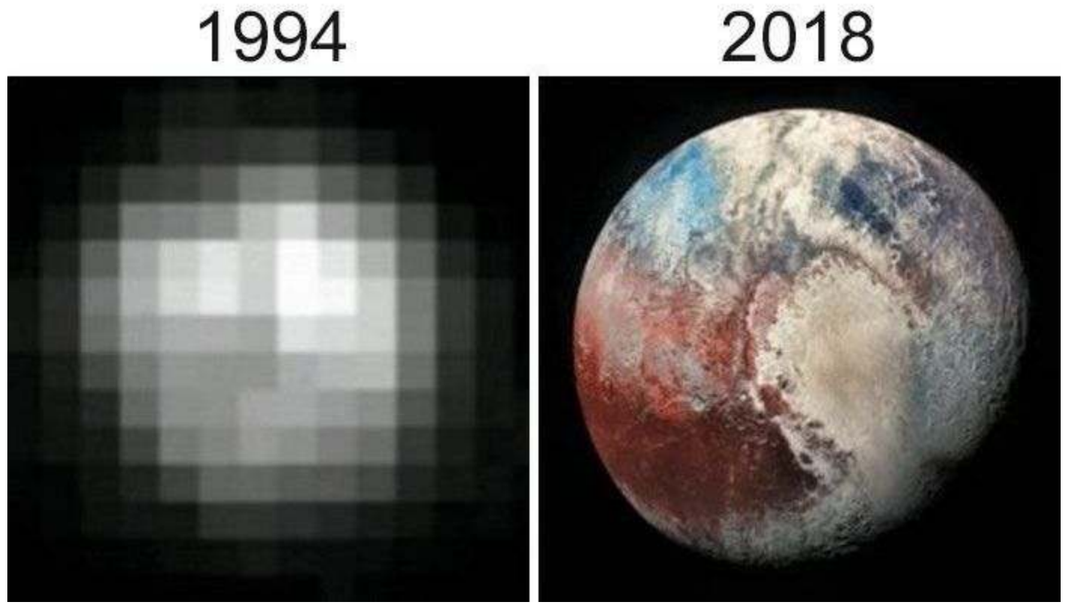
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Getting measurement right



Getting measurement right



Pluto did not change between these two pictures. **The measurement did.**

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From concepts to indicators

Conceptualization, operationalization, scoring (Adcock & Collier)

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Practice

Four short exercises today; simulating measurement error in Lab 4

From concepts to indicators

"Researchers routinely make complex choices about linking concepts to observations, that is, about connecting ideas with facts. These choices raise the basic question of measurement validity: **Do the observations meaningfully capture the ideas contained in the concepts?**"

Adcock & Collier (2001), *APSR*

Four levels, three tasks

Adcock & Collier (2001), *APSR* 95(3)

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Background concept

The broad, contested meanings of a term

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conceptualization

Systematized concept

The explicit definition you adopt

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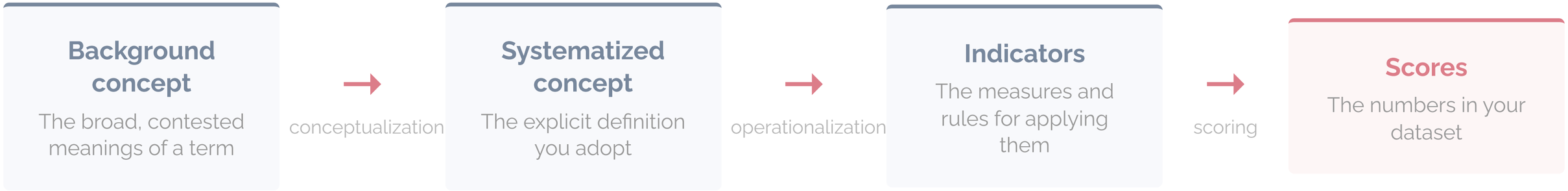
Indicators
The measures and rules for applying them

→ scoring

Scores
The numbers in your dataset

Adcock & Collier (2001), *APSR* 95(3)

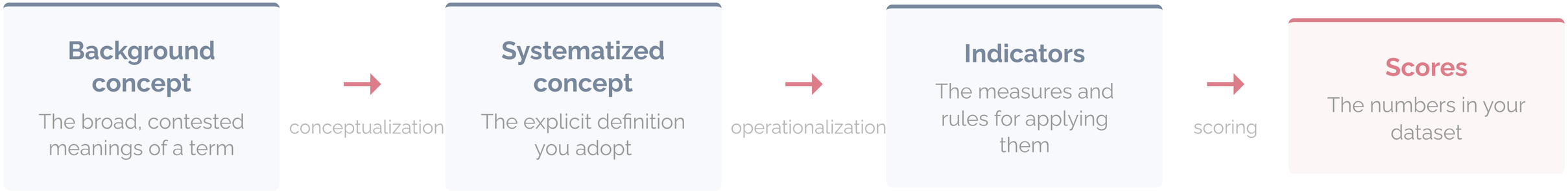
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- Measurement validity is about each **link** in this chain: do the scores capture the systematized concept you say they do?

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Four levels, three tasks



- Measurement validity is about each **link** in this chain: do the scores capture the systematized concept you say they do?
- A critique of a measure is always a critique of **a specific link**. Say which one.

Adcock & Collier (2001), *APSR* 95(3)

“Bike-friendly,” four ways

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A city can rank first on supply and last on use. **Which one is “bike-friendly” depends on the systematized concept.** Adcock & Collier make the same point with *democracy*.

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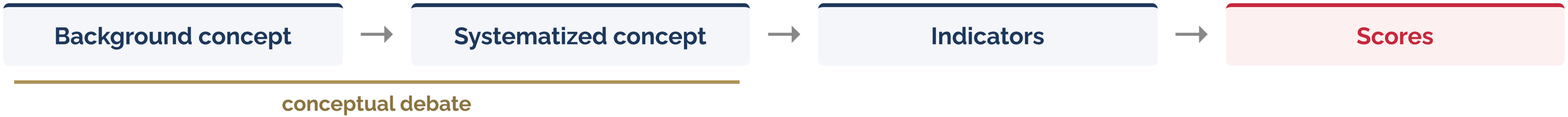
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Example: one study finds that protected lanes “make cities bike-friendly” (measured as perceived safety); another finds “no effect on bike-friendliness” (measured as cycling share after one year). Both can be right. The disagreement is conceptual.

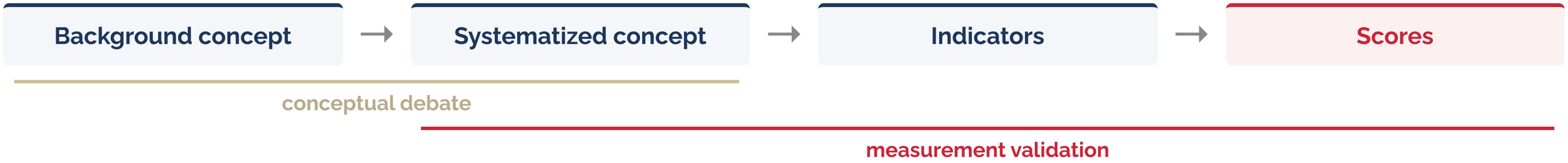
Validation targets the systematized concept

Background concept → Systematized concept → Indicators → **Scores**

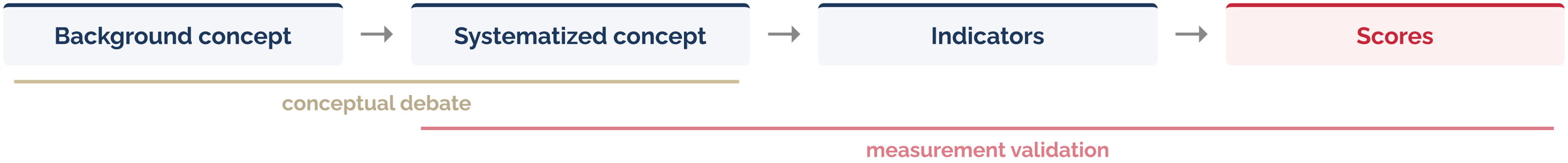
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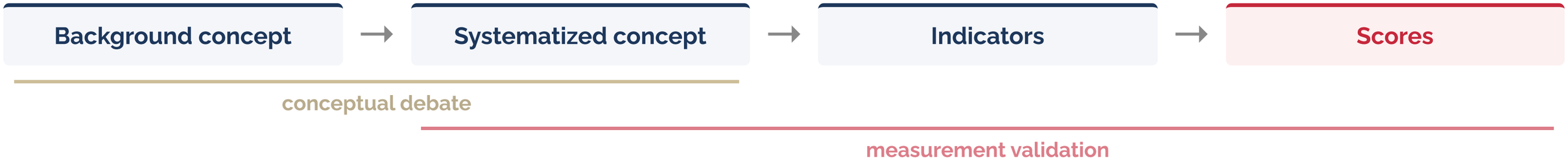


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- **Validity**: do the scores meaningfully capture the systematized concept? **Validation**: the evidence and arguments we use to assess that.

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- **Validity**: do the scores meaningfully capture the systematized concept? **Validation**: the evidence and arguments we use to assess that.
- Adcock & Collier: validation should focus on the link from the **systematized** concept to the scores. Whether the systematized concept does justice to the background concept is a separate, conceptual question.

How ‘democratic’ is the U.S. out of 10?

Adcock & Collier (2001), *APSR* 95(3)

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Before you criticize a score, **say which question you are asking**.

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Your turn ✏️ (~7 min, in pairs, then swap)

Pick a concept from your own **research area**. Write down:

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Swap with the pair next to you: find one case their indicators would misclassify. Is your objection a measurement concern or a conceptual one?

Validity is an argument

Different kinds of validity, different questions

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WEEK 3

Thinking causally (mostly internal)

Does the comparison identify the causal effect in this study?

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TODAY, PARTS 1–3

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- **Measurement validity** matters most for concepts we cannot observe directly: attitudes (prejudice, polarization), traits (political knowledge), macro concepts (democracy, state capacity).
- How do we argue that a measure is valid? With **evidence of different kinds**, which different fields name differently.

Content, criterion, construct

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Three names for three kinds of **evidence**. Are they really three kinds of validity?

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> The labels are **field-specific**. Political scientists say “content” and “construct validity”; psychometricians say “sources of evidence”; ML says “ground truth” and “benchmarks”. **Speak your discipline’s language**, but keep the idea underneath… you argue for an interpretation with evidence.

Five sources of validity evidence

Source of evidence	The question	Typical evidence	In Adcock & Collier

AERA, APA & NCME (2014), *Standards for Educational and Psychological Testing*

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The evidence comes in different kinds. **Let's look at some.**

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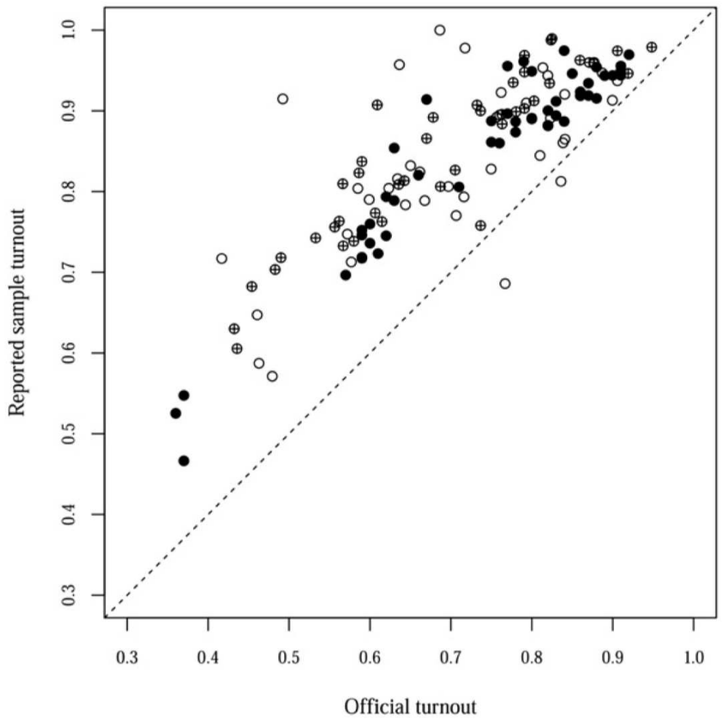


Fig. 1. Reported sample turnout rates from 130 postelection surveys versus official turnout. Data are taken from Modules 1–3 of the *Comparative Study of Electoral Systems (CSES)*, and from a collection of election surveys for which vote validation studies (VVS) are available. For more detailed information, see the Appendix to this paper.

Selb & Munzert (2013), *Electoral Studies* 32(1)

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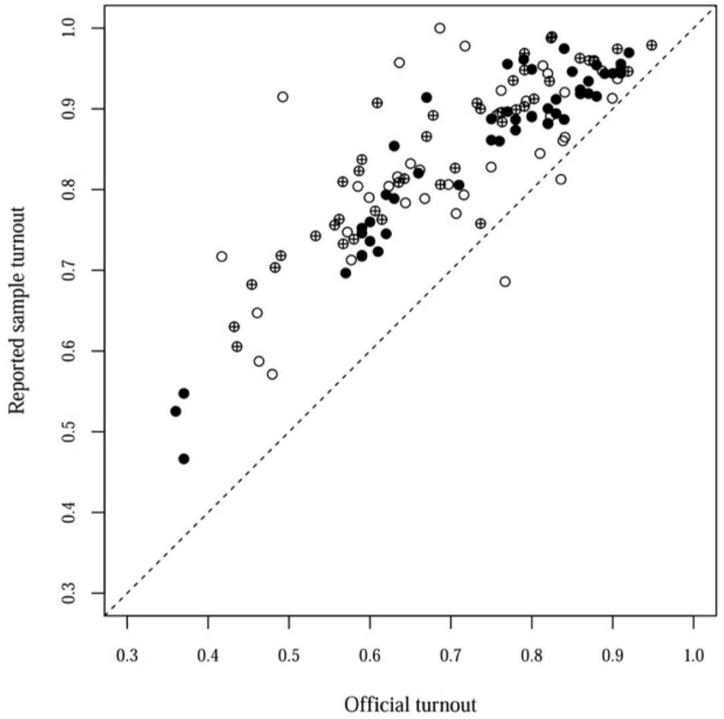


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- Across 130 post-election surveys, **reported turnout is almost always higher than official turnout**.

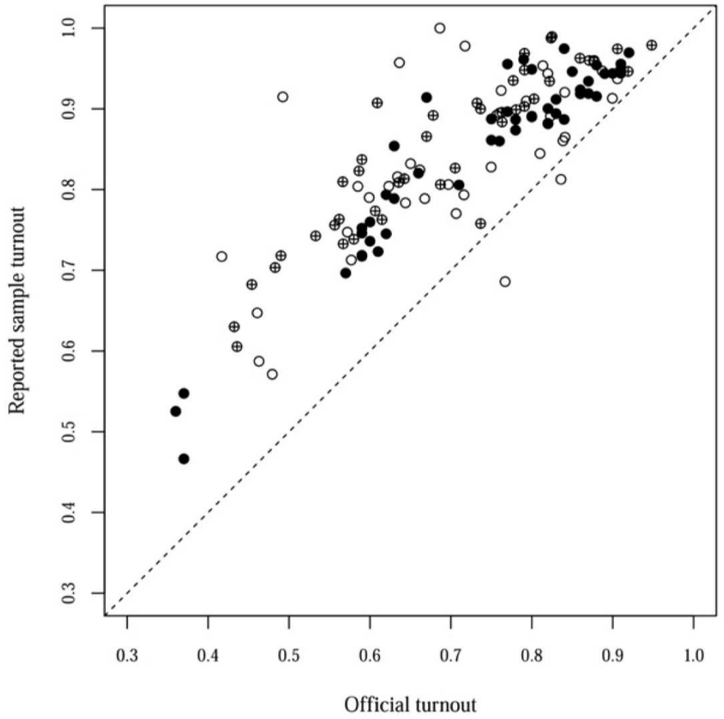


Fig. 1. Reported sample turnout rates from 130 postelection surveys versus official turnout. Data are taken from Modules 1–3 of the *Comparative Study of Electoral Systems (CSES)*, and from a collection of election surveys for which vote validation studies (VVS) are available. For more detailed information, see the Appendix to this paper.

Selb & Munzert (2013), *Electoral Studies* 32(1)

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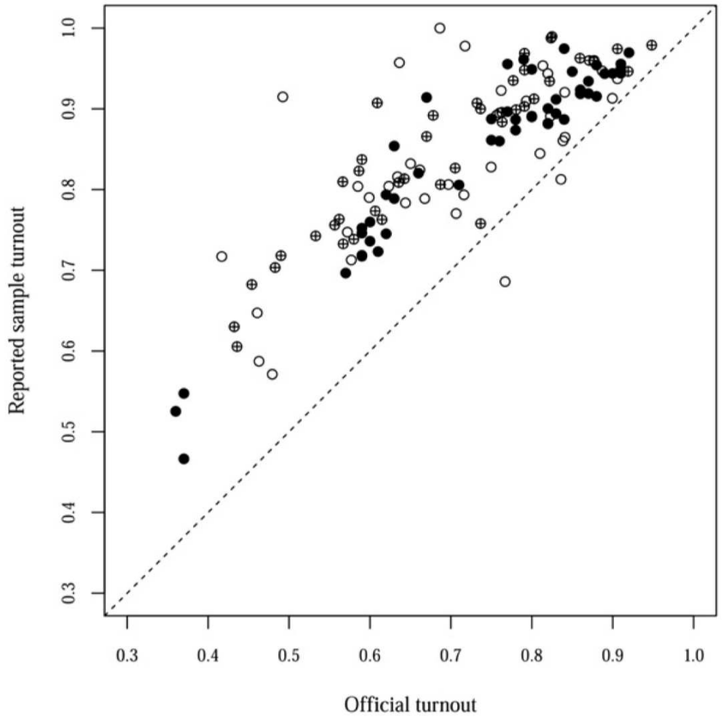


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- An "obvious" item can be **systematically** wrong.

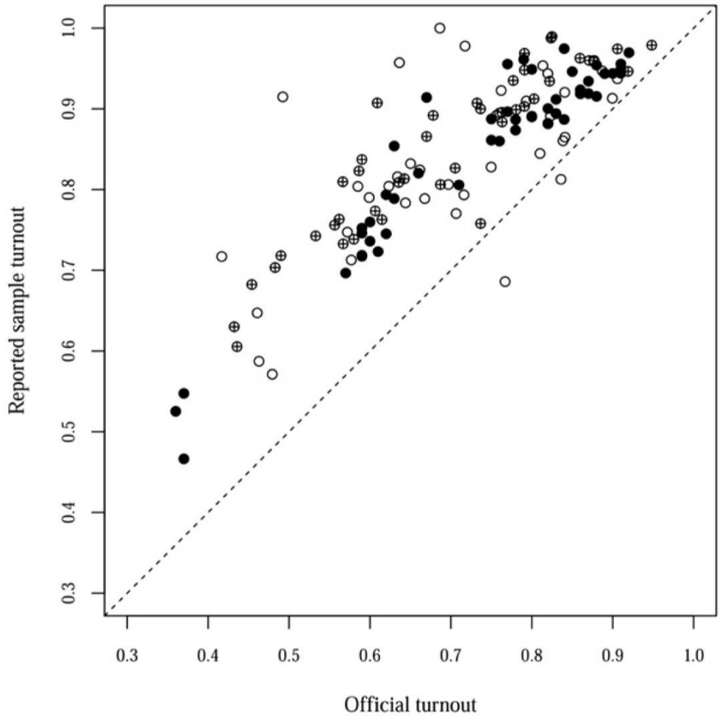


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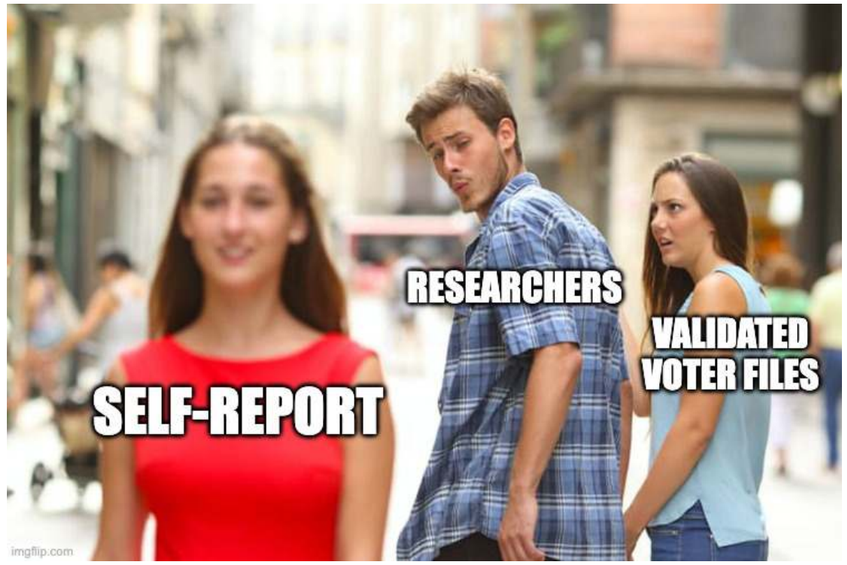
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Example: affective polarization is often measured by asking how warmly people feel toward “the Republican Party” or “Democrats”. When Druckman & Levendusky (2019) asked who people have in mind, most thought of **politicians, not ordinary voters**, and they disliked politicians far more.

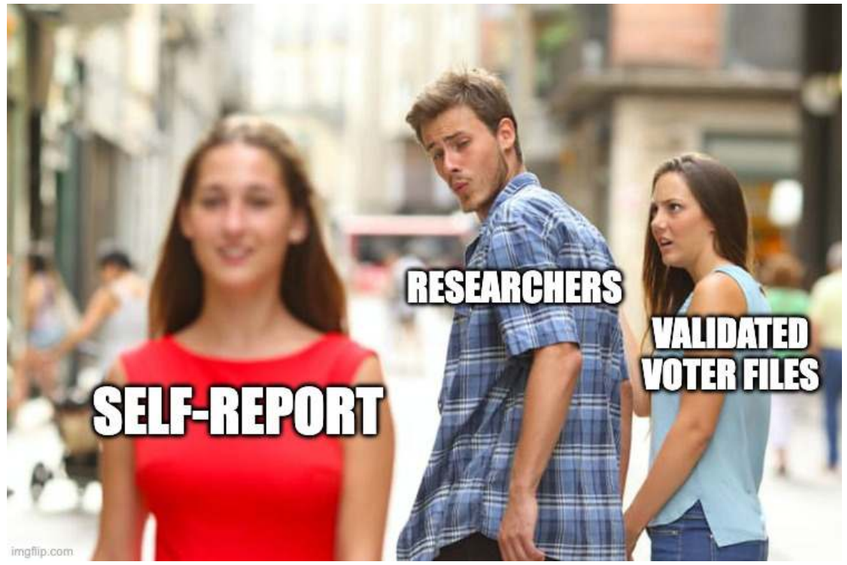
The thermometer partly measures dislike of party elites, not of fellow citizens.

Criterion validity: concurrent and predictive



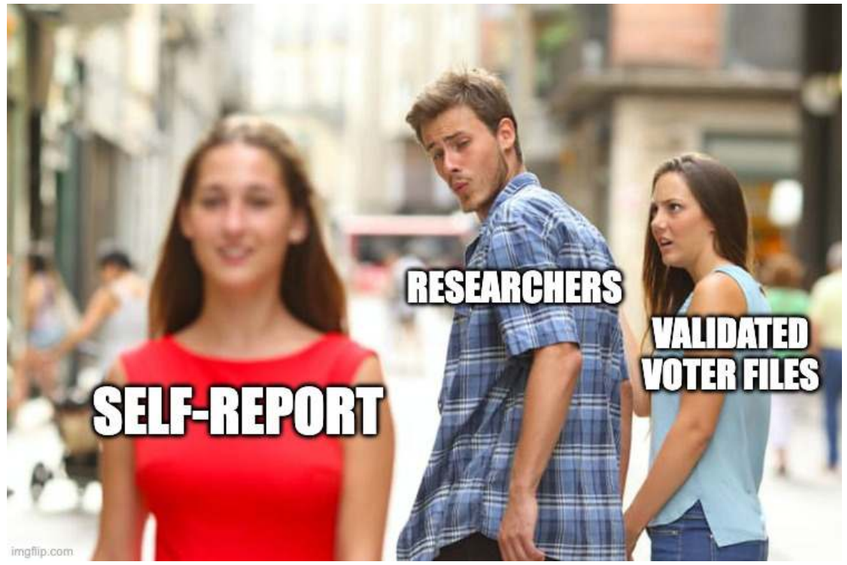
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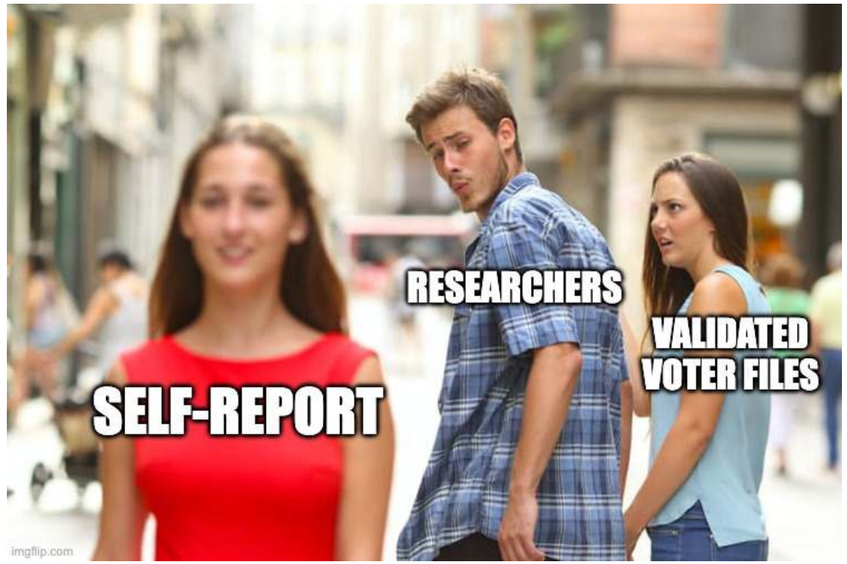
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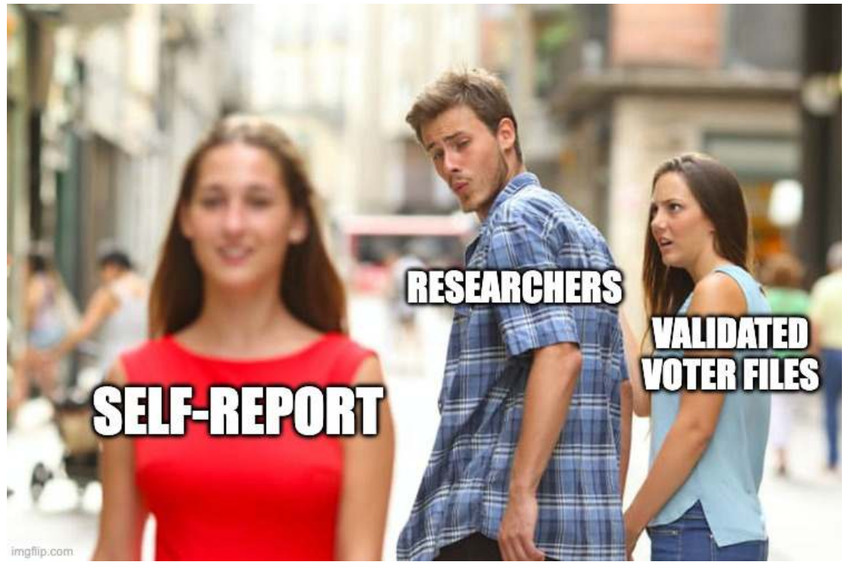
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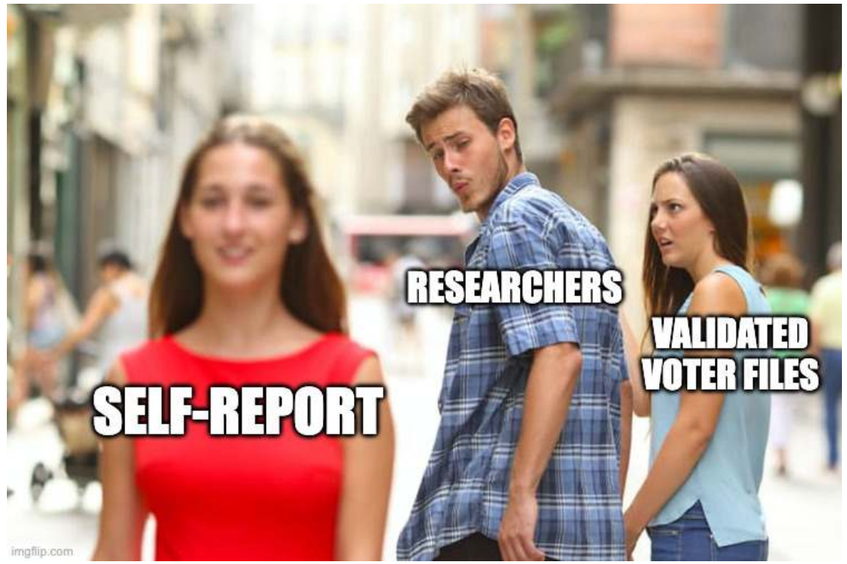
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- **The dilemma:** if a gold standard exists, why build a new measure? If none exists, **what is the benchmark?**

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- In sociology: law schools reorganized admissions, spending, and how they report job placement around the *U.S. News* rankings (Espeland & Sauder 2007). Measures are **reactive**: they can change the thing they rate.



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One measure, all the evidence: digital literacy

Guess & Munger (2023) define digital literacy as **online information discernment combined with the basic digital skills necessary to attain it**, and measure the skills part.

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1. **Content**: three candidate scales

Internet skills (Hargittai)
How familiar are you with terms like "hashtag", "wiki", "phishing", "spyware"?

Power user (Sundar & Marathe)
"I make good use of most of the features available in any technological device."

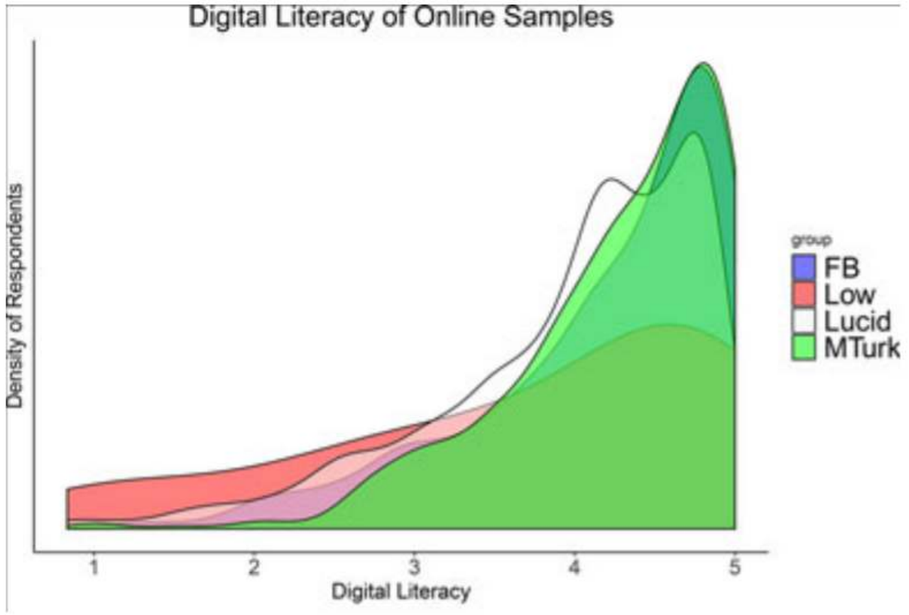
"Low end" (new)
Five items built to pick out the least skilled users, the ones most studies of misinformation care about.

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Share below 2.5 of 5 on internet skills: tech workers **0%**, MTurk 1.4%, Facebook ads 5.1%, Lucid 5.8%, library computer class **16.7%**. Paper's Figure 2.

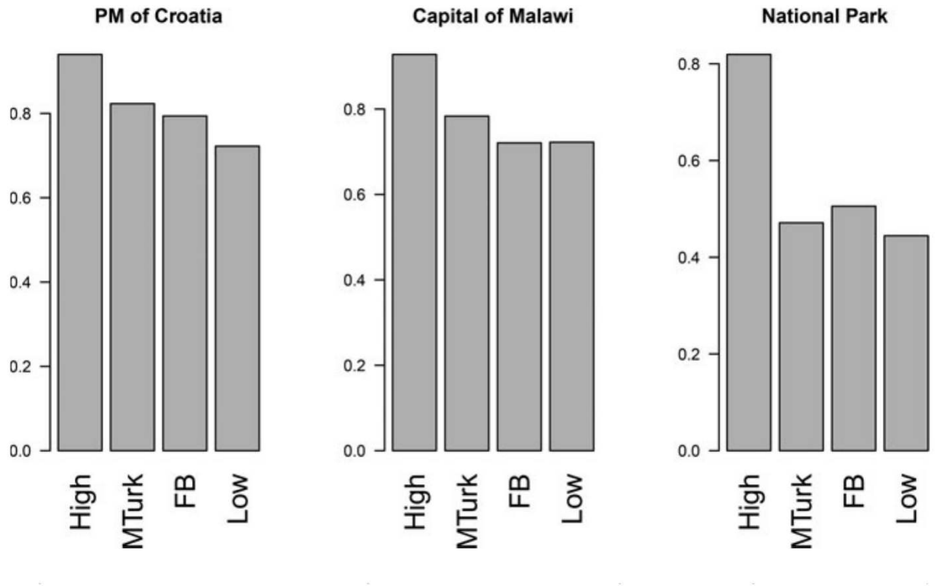
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Told to **look it up**: Who is the Prime Minister of Croatia? What is the capital of Malawi? Which is the only US national park that begins with "T"?



Share correct by sample (paper's Figure 1). Tasks with **known correct answers** are a criterion.

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4. **Why it matters**: age, samples, and time

Age

Older users share more misinformation. Is it age, or digital literacy that age stands in for? A confounder you can only adjust for if you can measure it.

Samples

Online panels like MTurk under-represent low-skill users: selection on the very trait you study.

Time

Items like "hashtag" age with the technology. Items without proper nouns may have better temporal validity.

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A good starting point for your Ph.Ds: Ask valid for what, for whom, and when?

Your turn ✏️ (~6 min, in pairs)

A team proposes a new 9-item scale of **affective polarization** toward voters (not politicians).
For **each** source, name one piece of evidence they should show.

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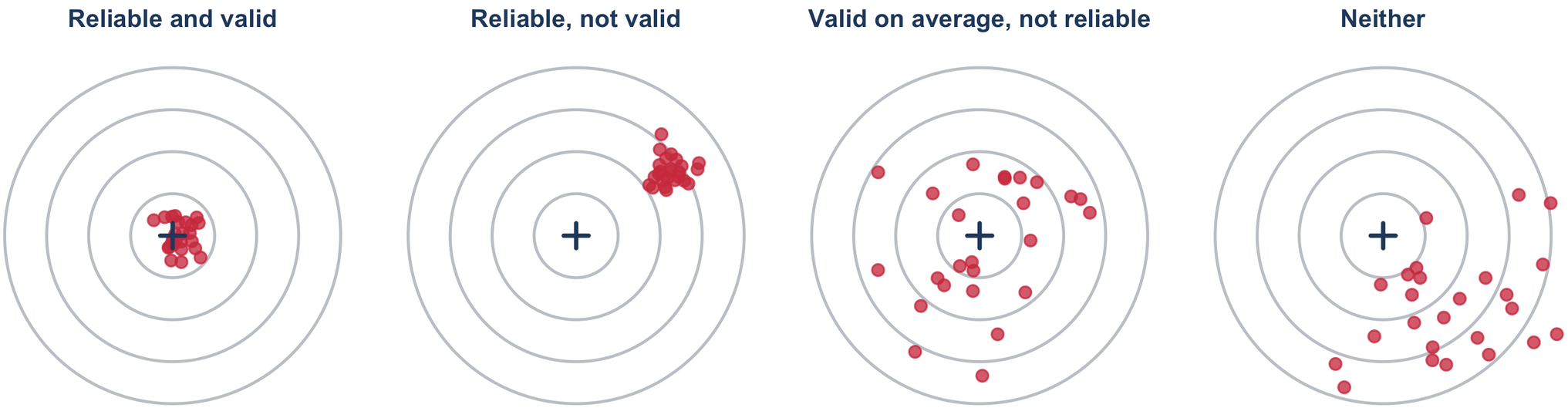
Which single empirical result would make you stop using the scale?

Break

10 minutes

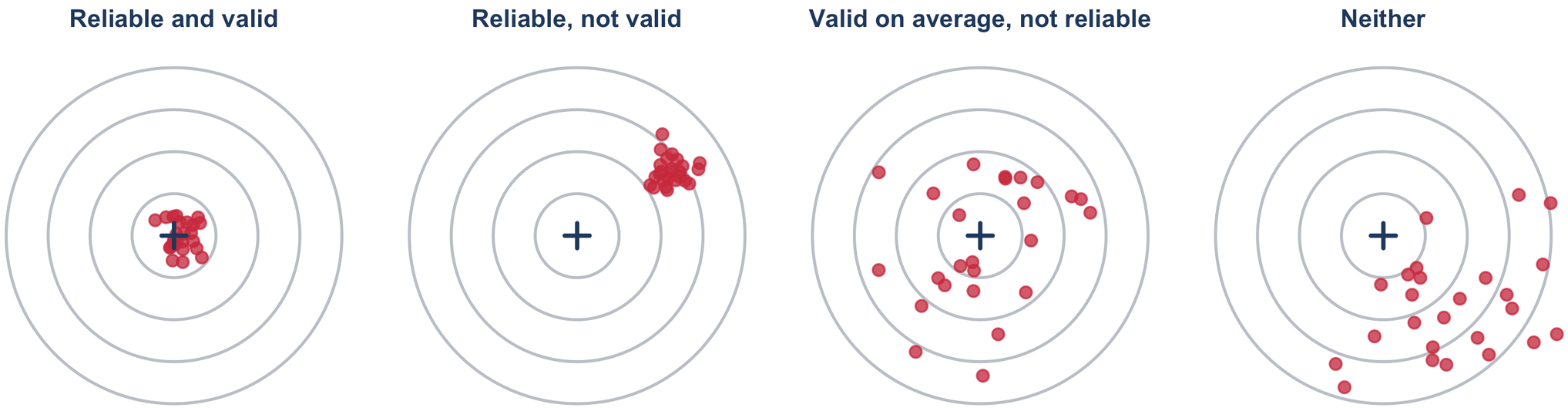
Reliability and measurement error

Valid is not the same as reliable



Each panel: 25 repeated measurements (red) of the same quantity. The navy cross is its true value.

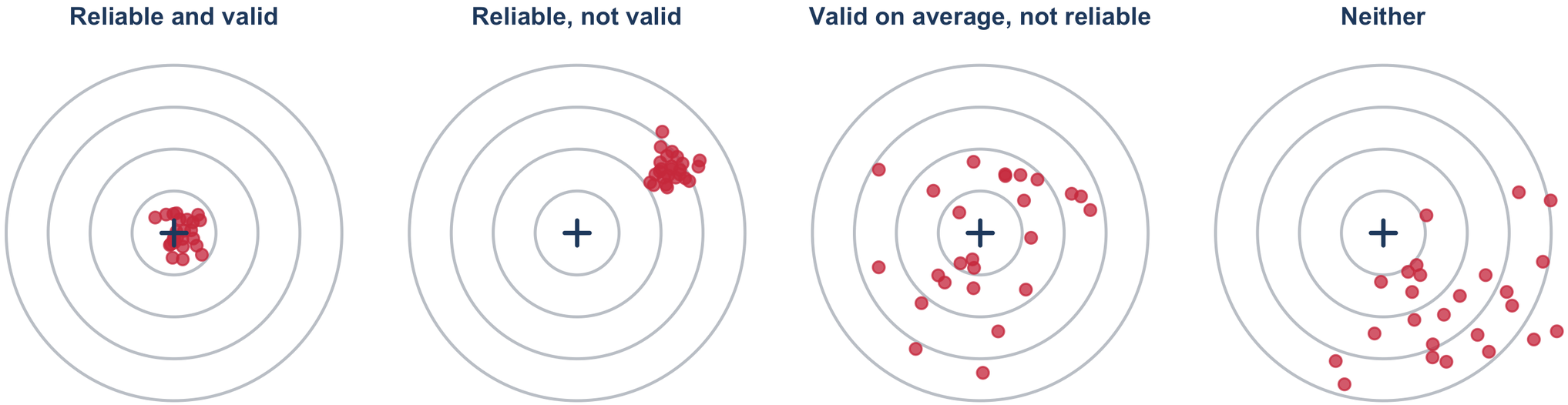
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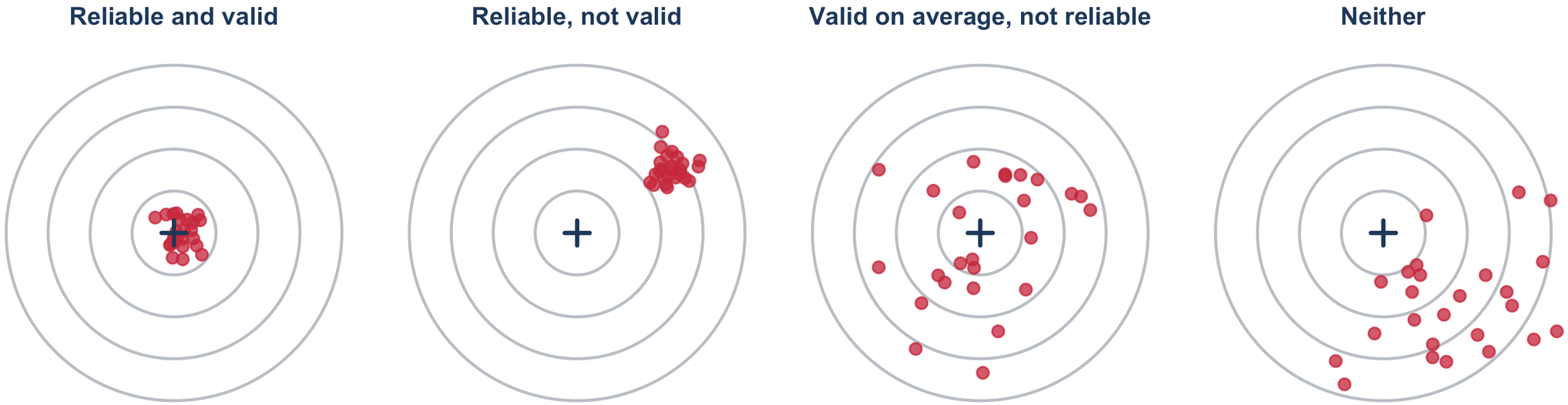
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- **Valid on average, not reliable**: a single survey question on an issue position. Right on average, but individual answers bounce between panel waves.

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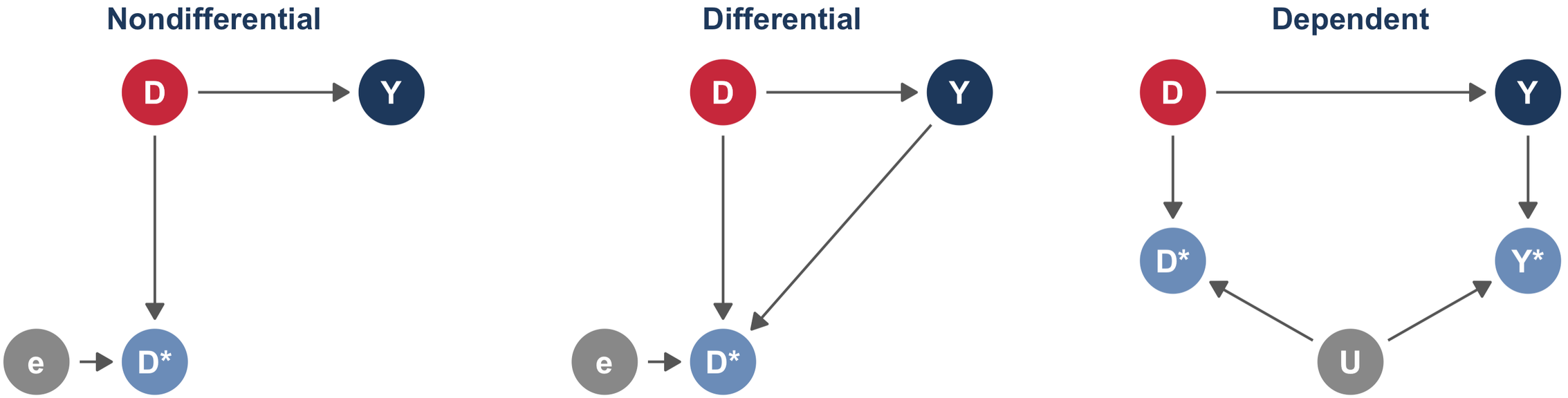
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Internal consistency

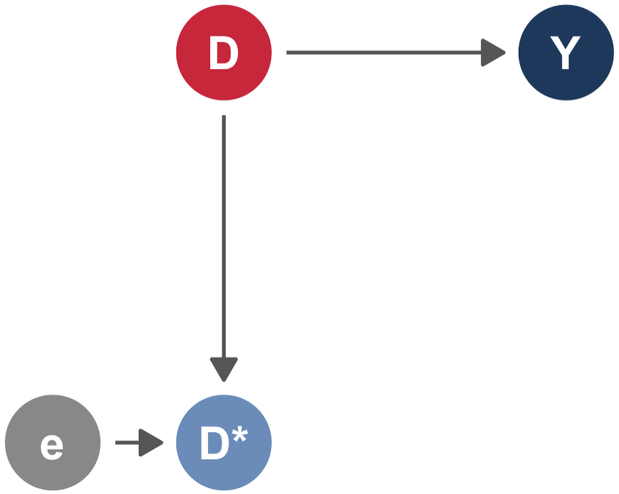
Several items for one construct, like a trust-in-institutions battery. Do they move together? (Cronbach's α, McDonald's ω: next week)

Measurement error is a DAG problem

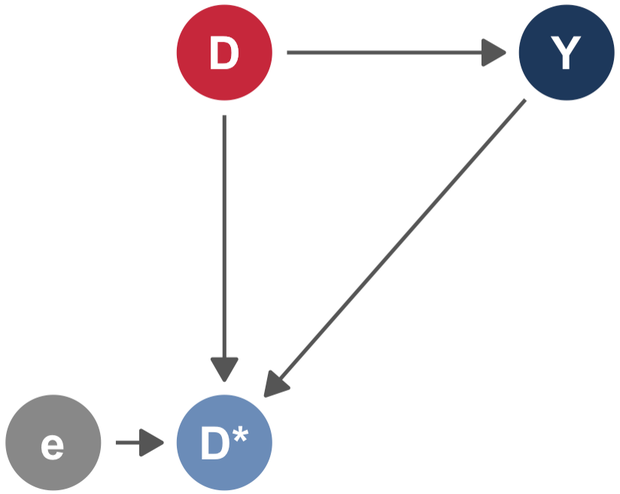


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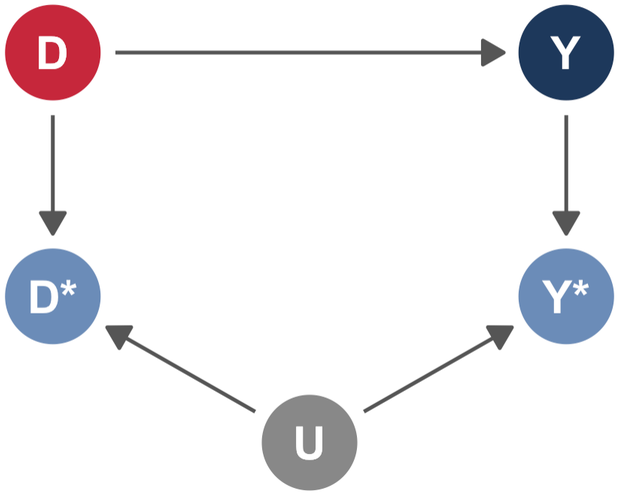
Nondifferential



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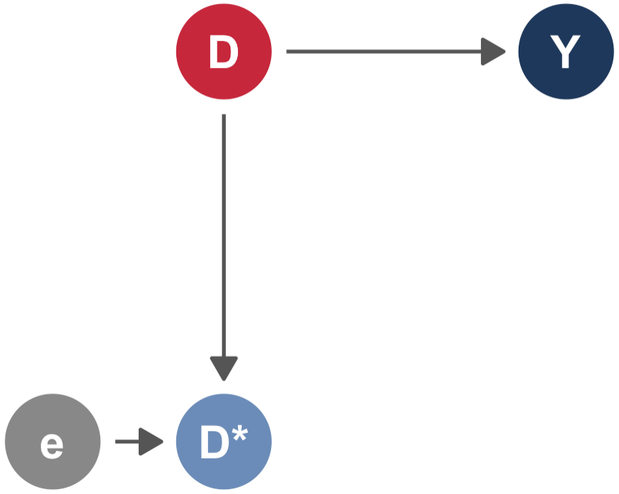
Dependent



Ad exposure (D) is recalled with random error, unrelated to whether people voted (Y).

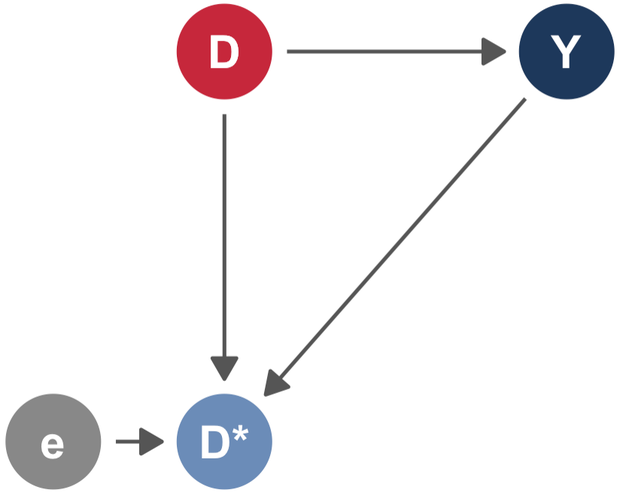
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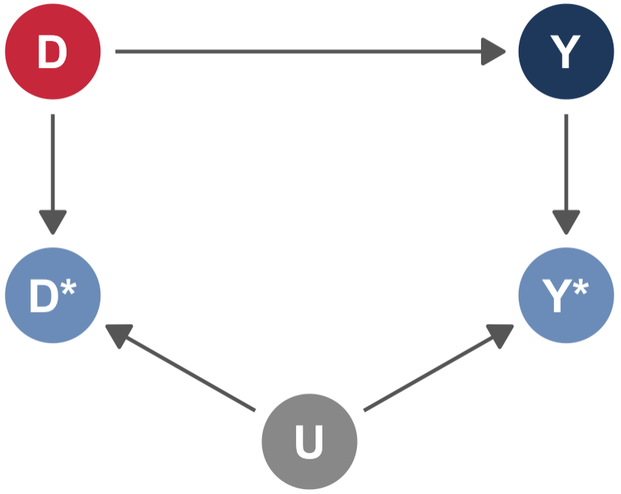
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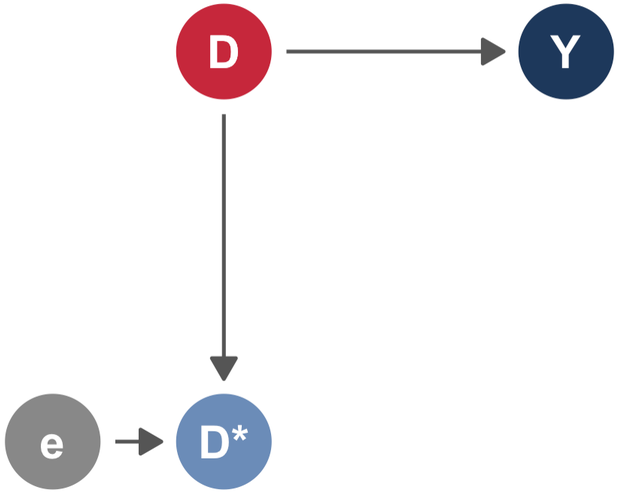
People who voted remember more of the campaign: the outcome feeds into the measured treatment.

Dependent



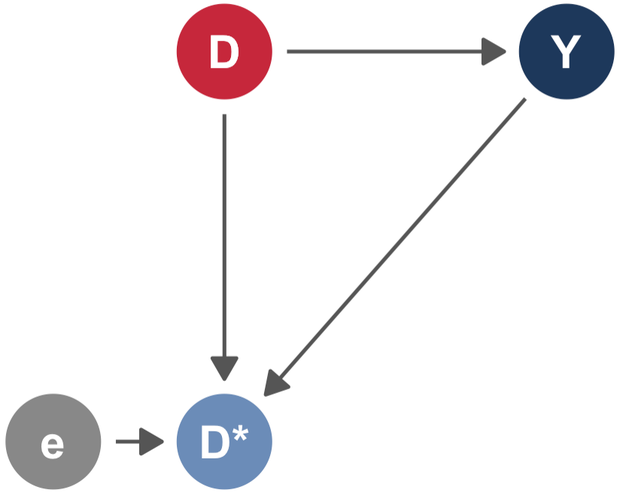
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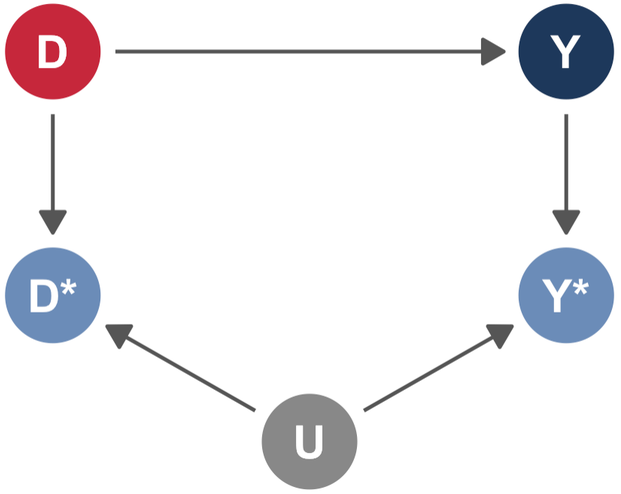
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Differential



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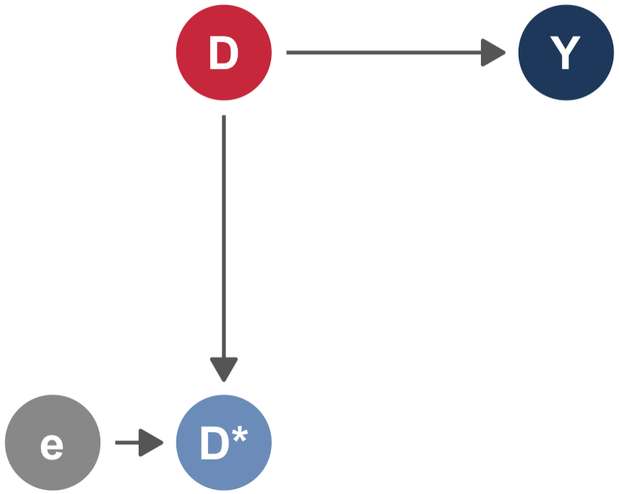
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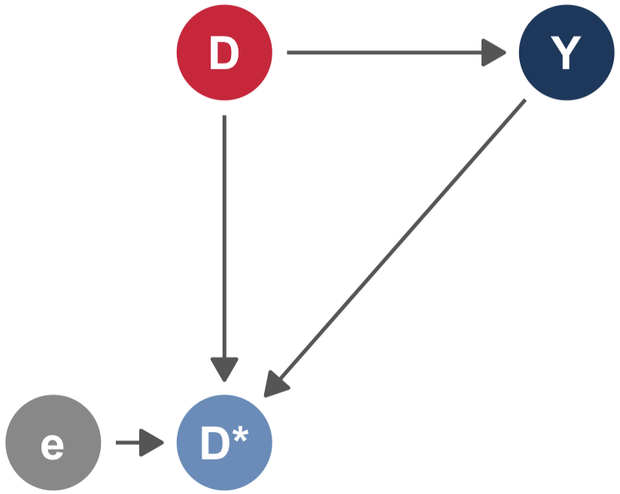
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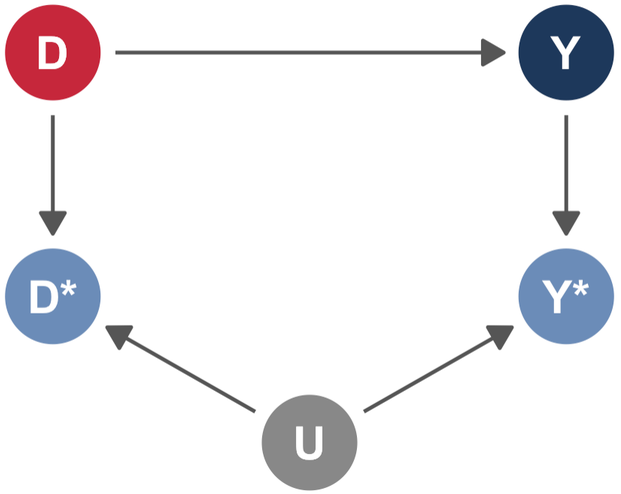
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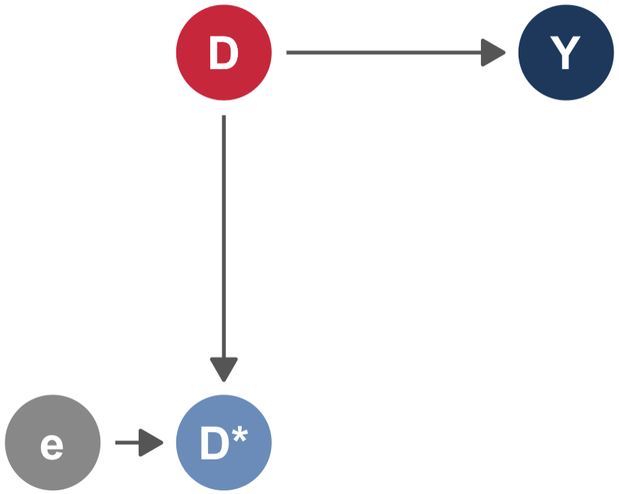


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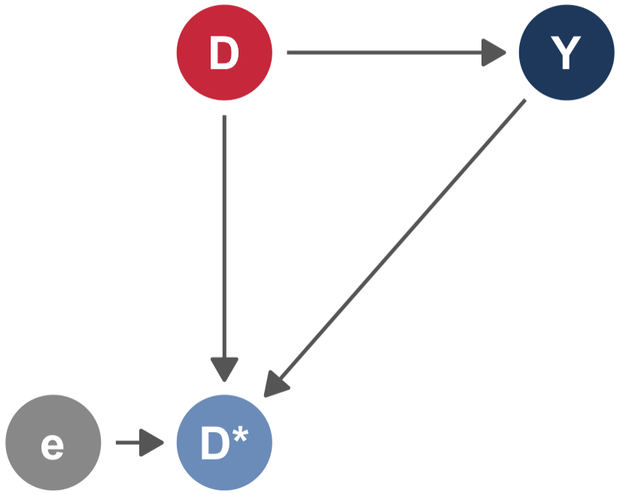
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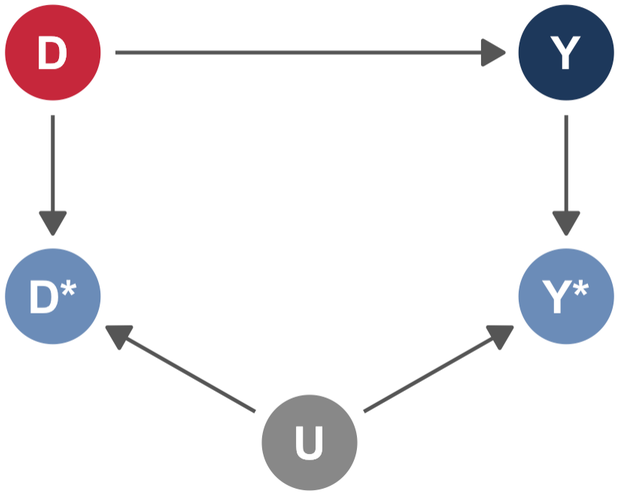
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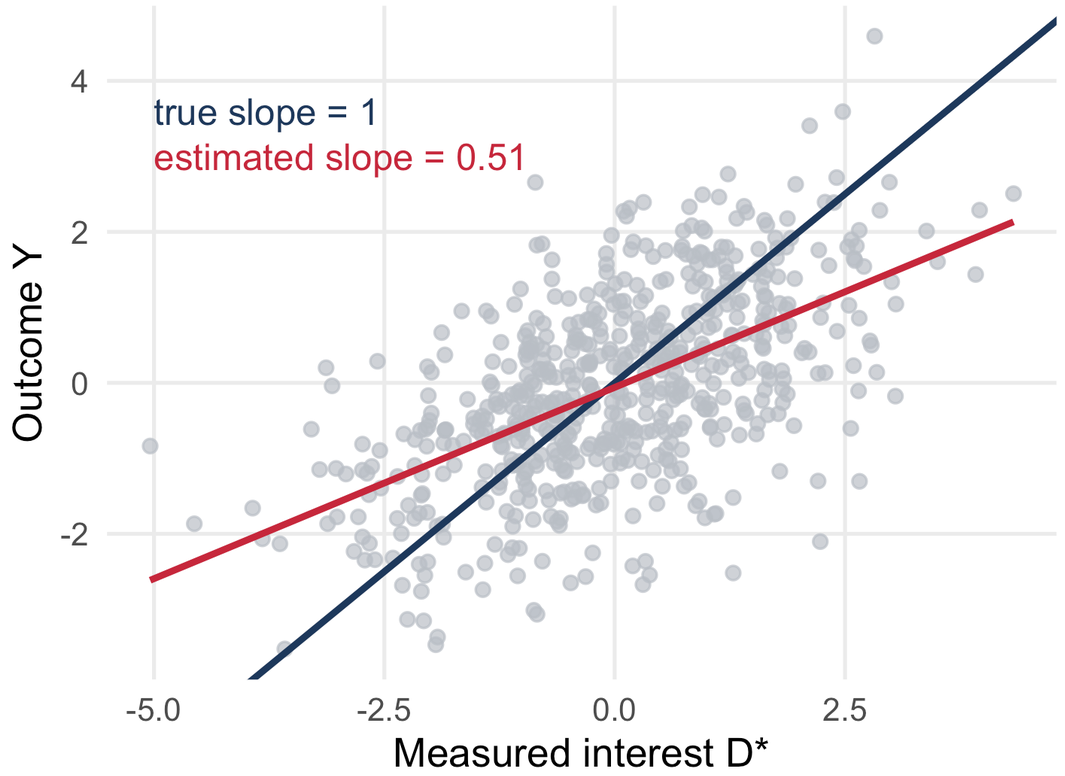
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- We analyze D^* and Y^*, not D and Y. The error is just **another node** in the DAG (Hernán & Cole 2009).
- The question is always: **which arrows point into the measured variables?**

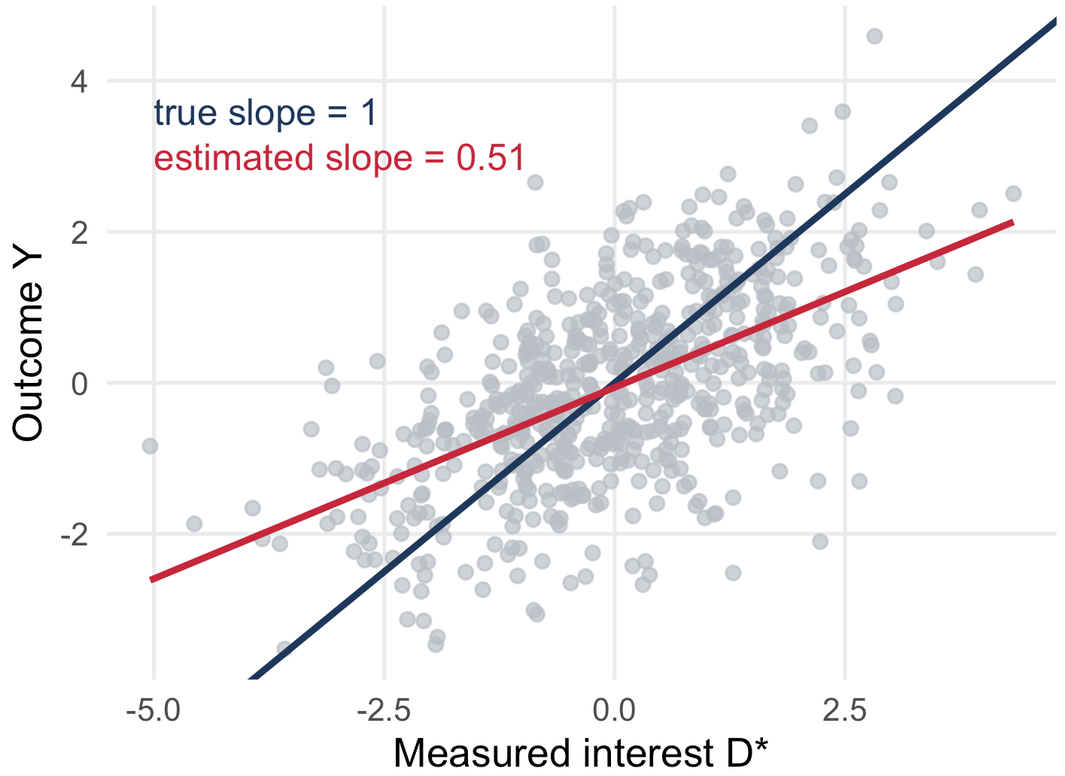
Error in the treatment: attenuation



Simulation: 600 respondents, true slope 1, interest measured with reliability 0.5. Navy: the true relationship. Red: what a regression on the noisy measure recovers.

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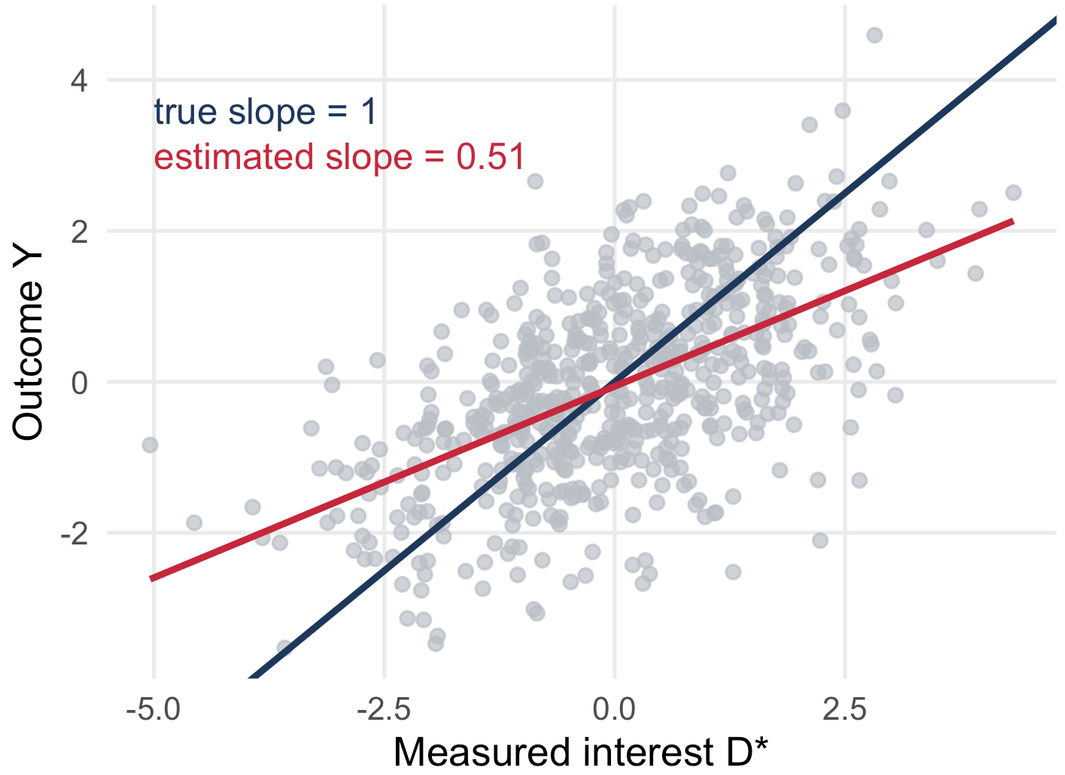


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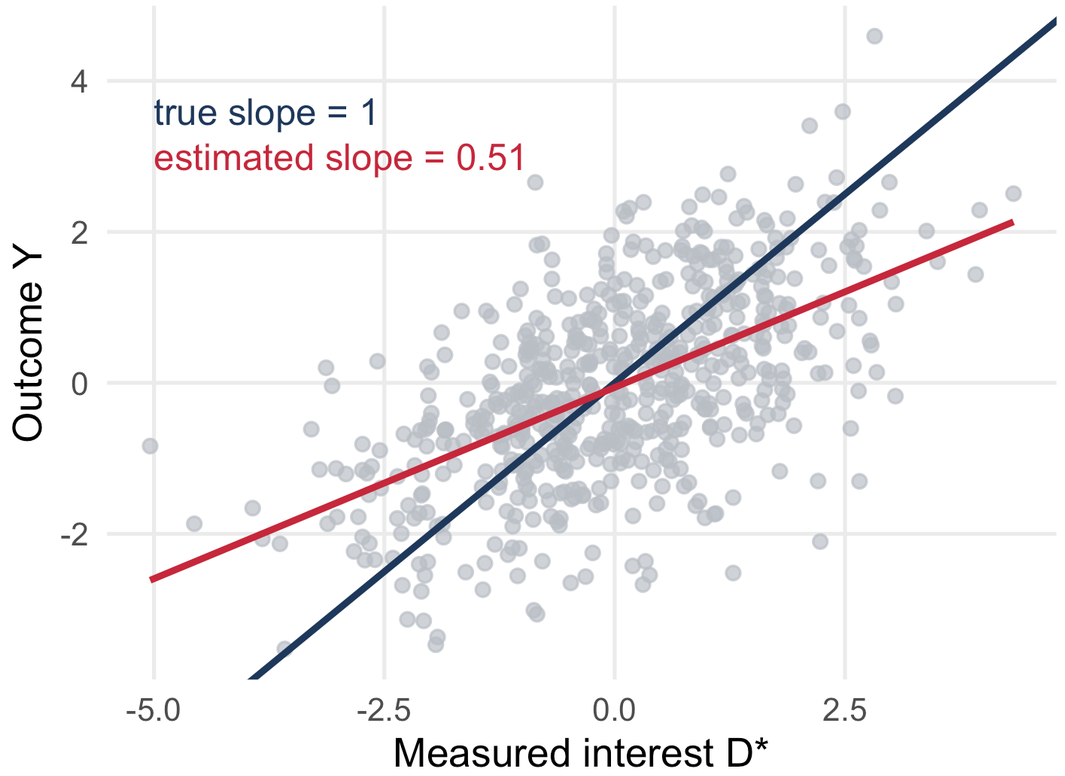
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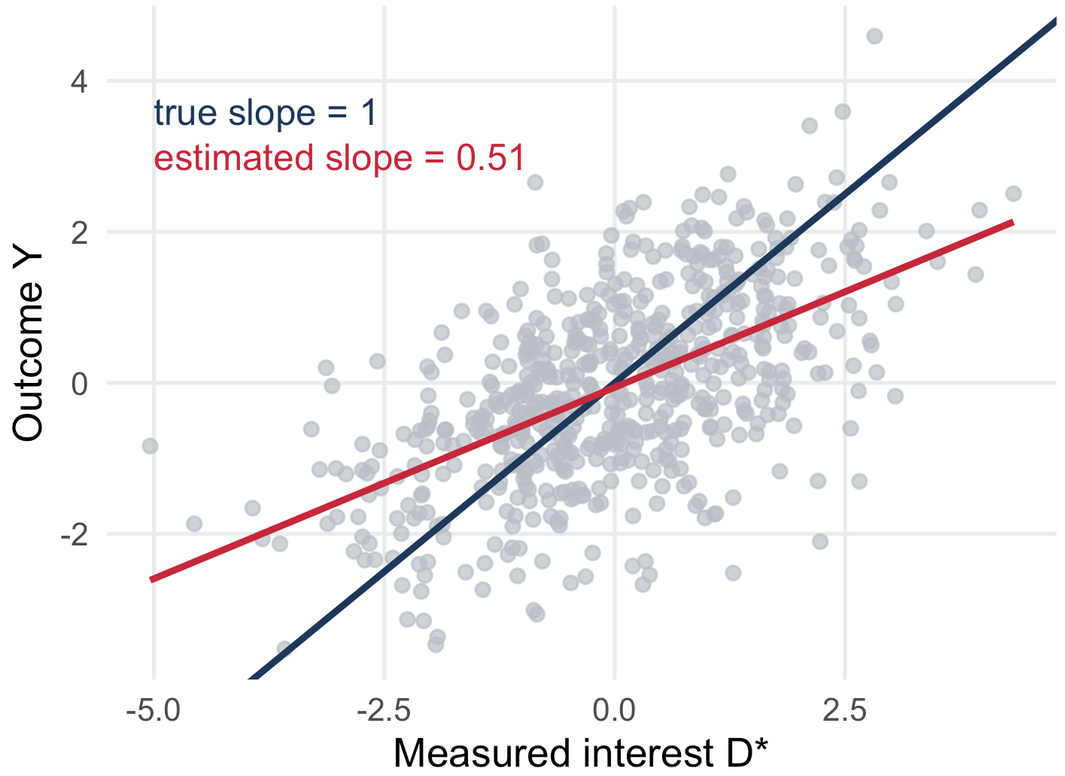
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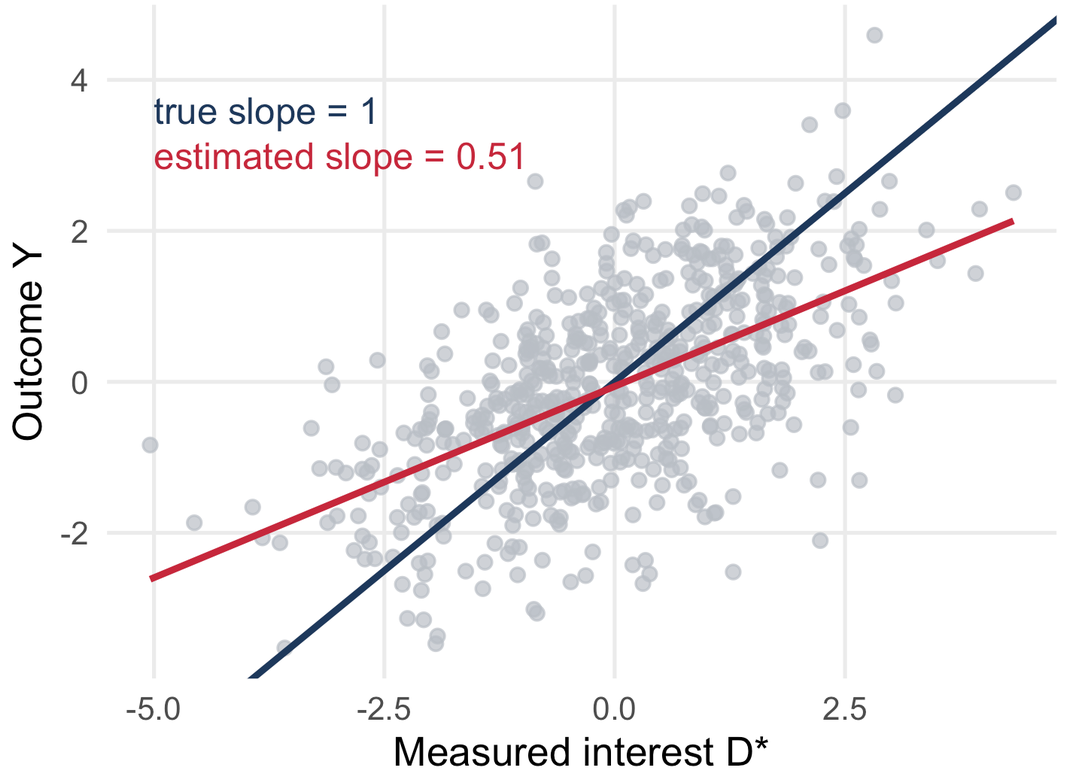
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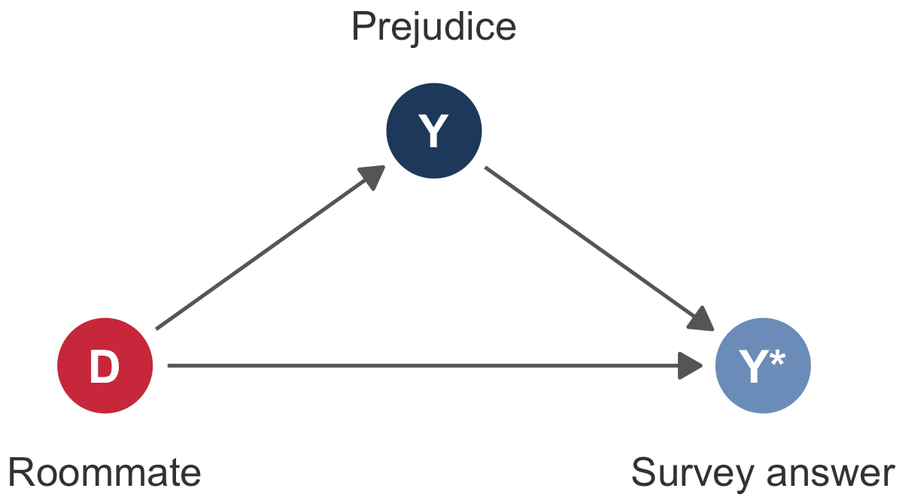
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- With $\rho = 0.5$, a true effect of 10 points shows up as 5: you conclude interest "barely matters".
- Horse races mislead: a well-measured rival (education) **looks more important**.



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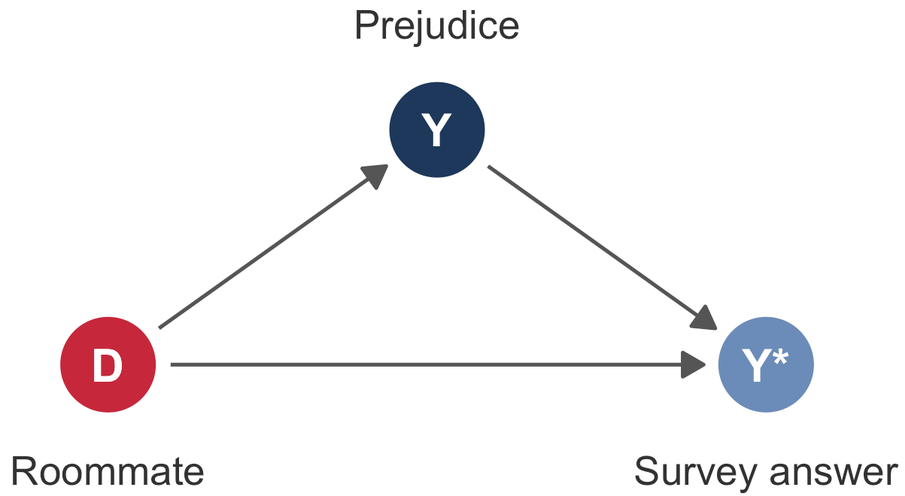
Error in the outcome: noise, unless it is differential



The arrow from Roommate to Survey answer is the problem: the treatment changes what people say, not only what they think.

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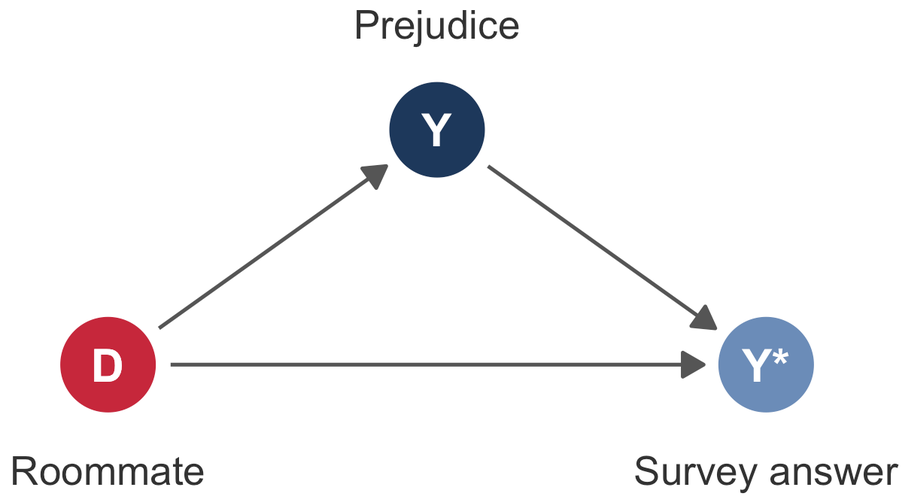
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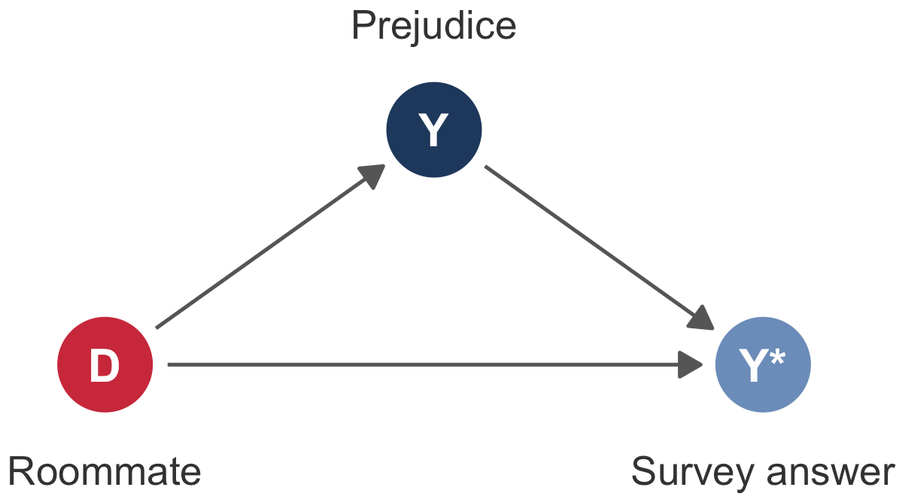
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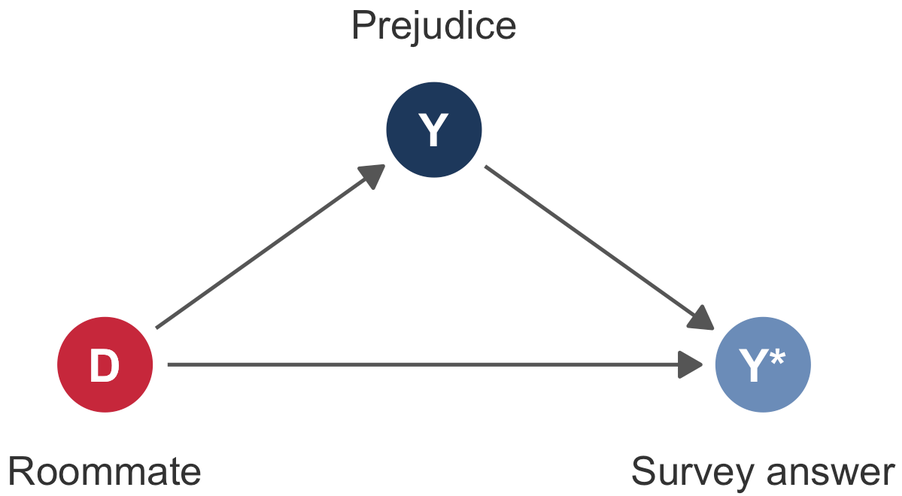
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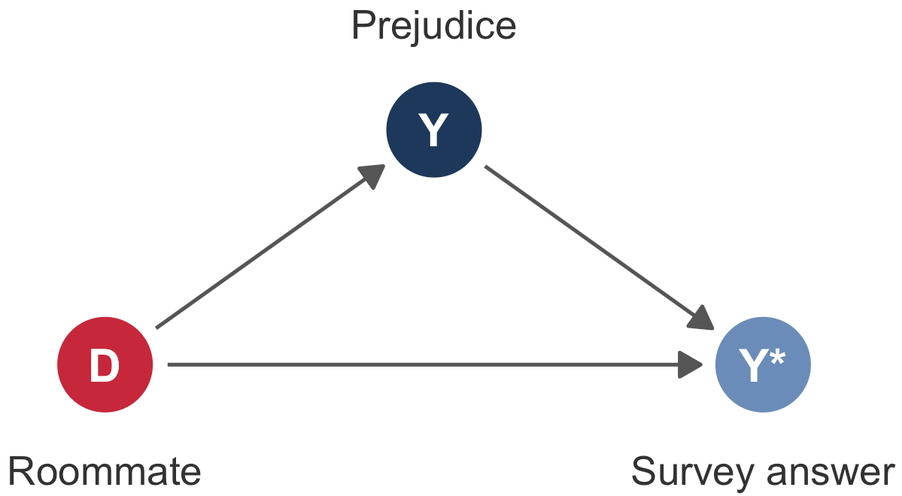
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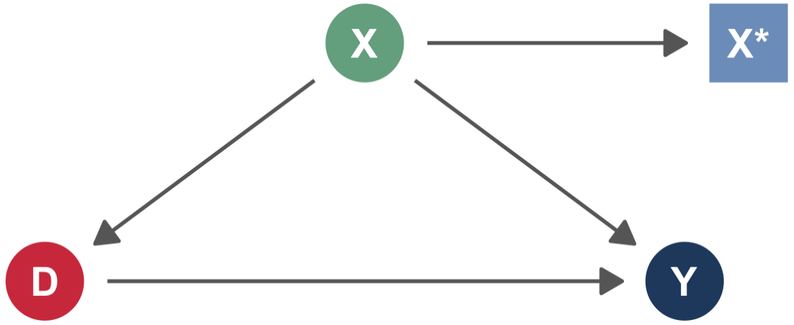
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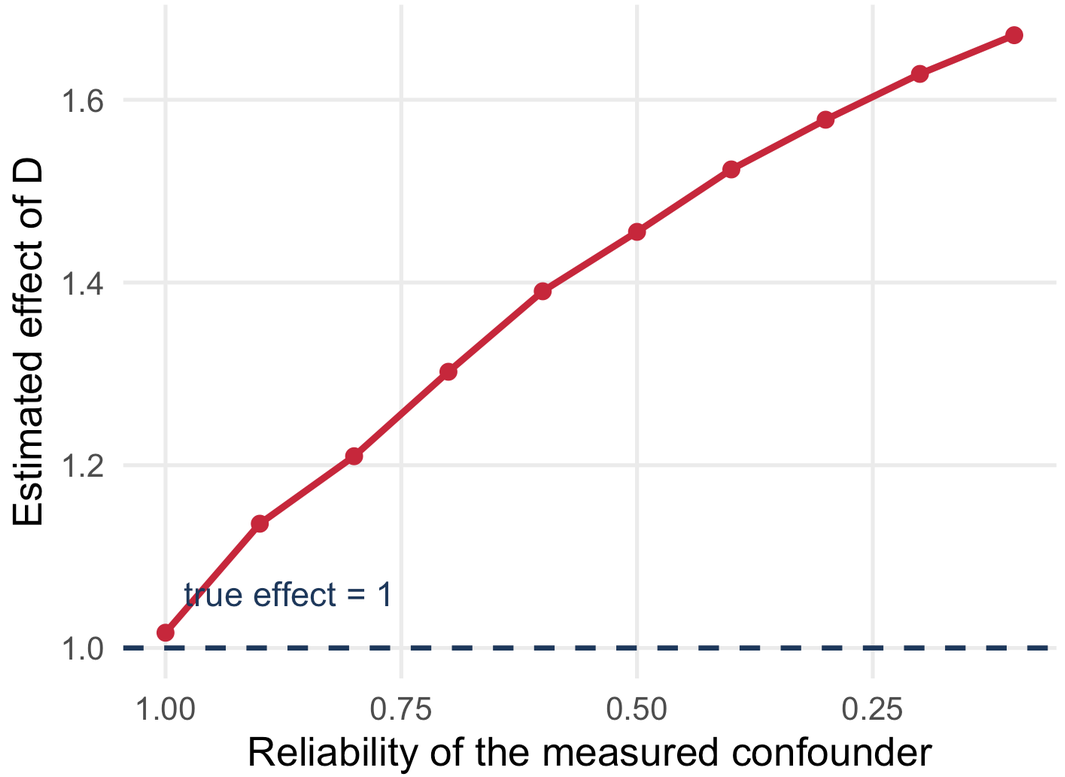
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- Part of the estimated "effect of contact" is an effect on **reporting**: the error is differential, and the bias can go in any direction.
- Same problem in any experiment with a **self-reported** outcome that the treatment makes socially desirable: turnout after a get-out-the-vote call, tolerance after a diversity training.



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Error in a control: residual confounding

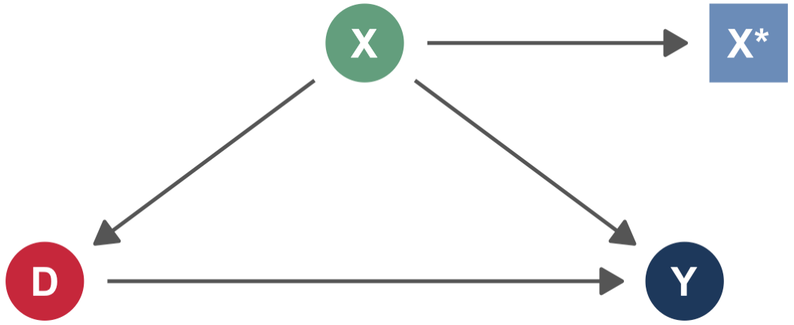


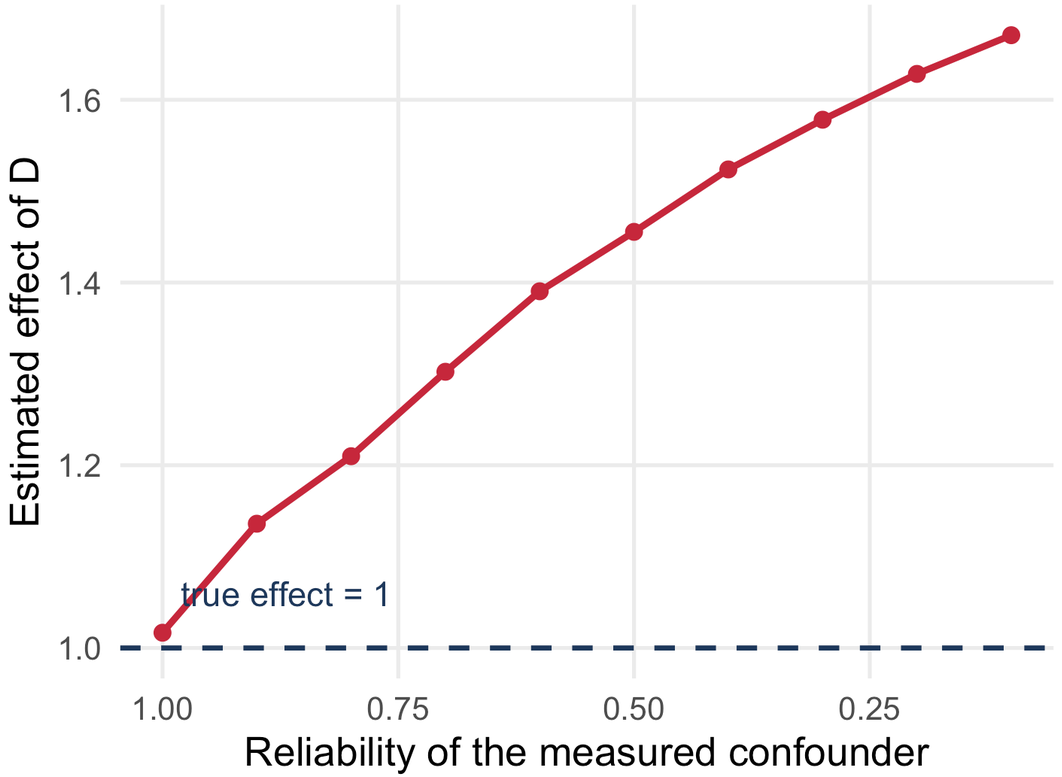


Simulation: we control for X* instead of X. With a perfect measure (left) the estimate is right; as the measure gets noisier (moving right), it drifts back toward the unadjusted, confounded estimate.

Error in a control: residual confounding

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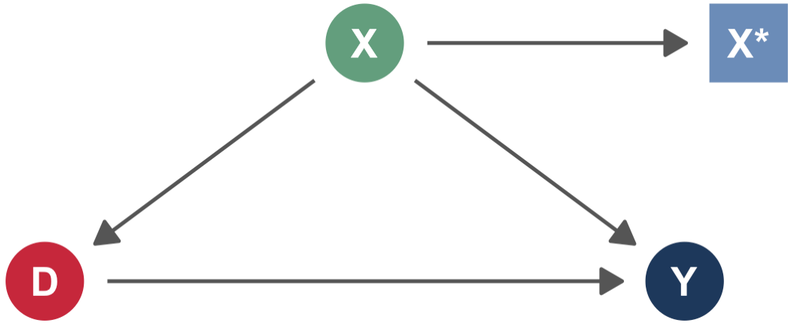




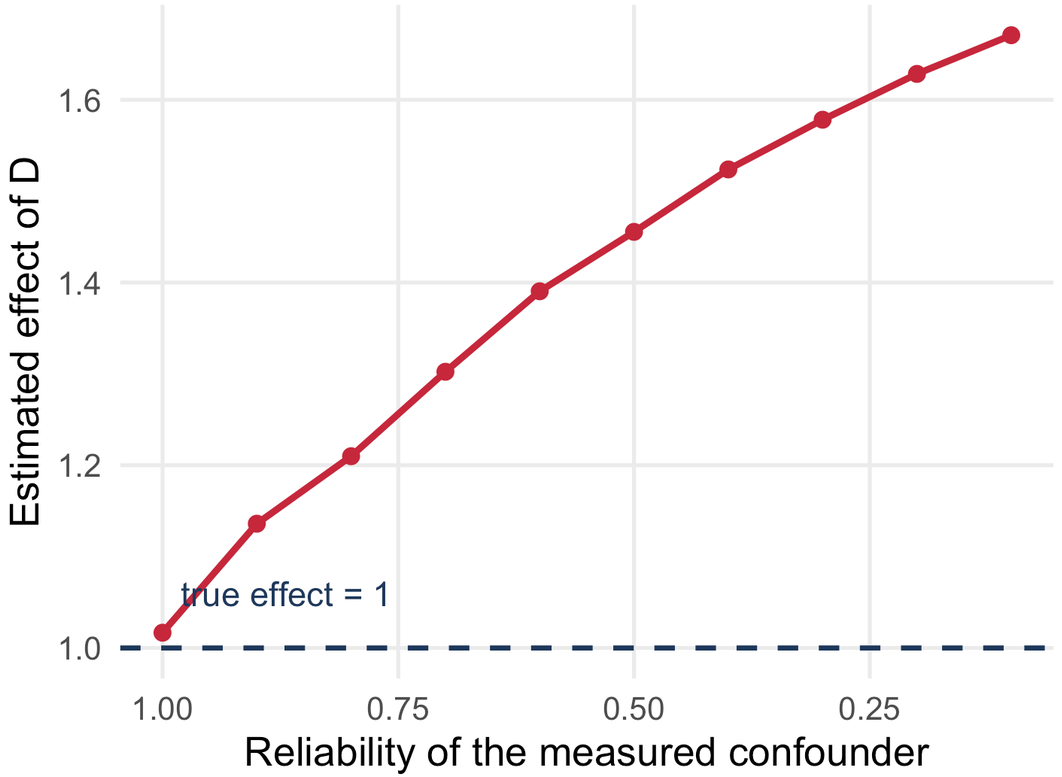
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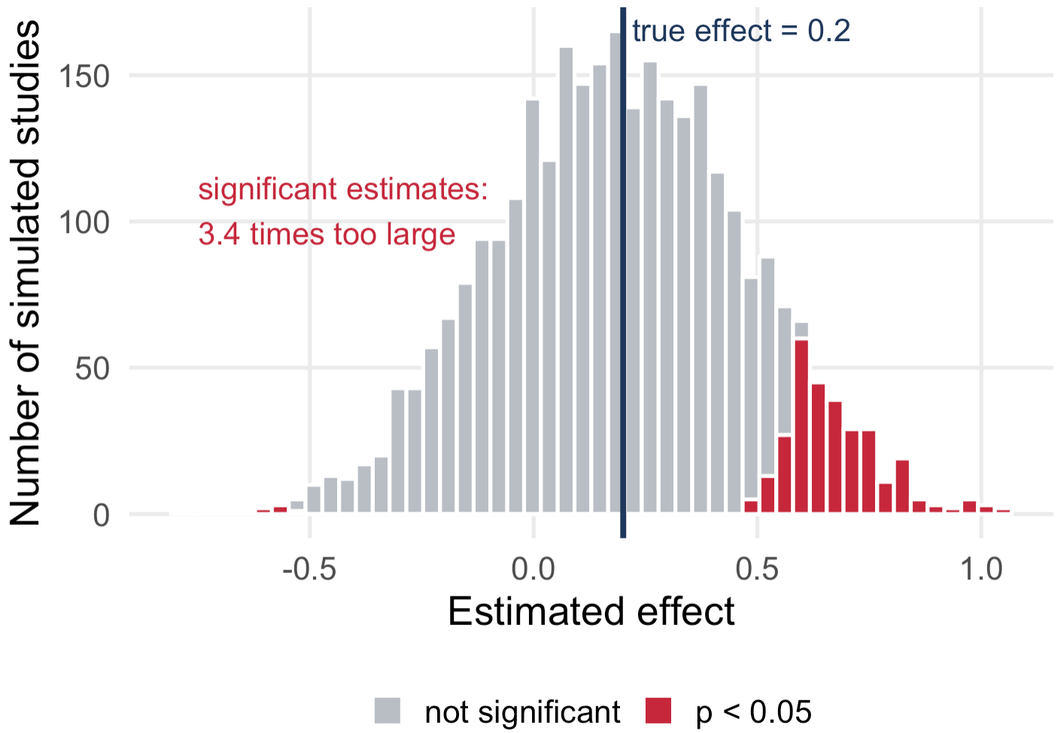


- Controlling for X^* closes the backdoor **only partly**: the leftover bias has the sign of the original confounding, and grows as reliability falls.



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Noisy measures and the replication crisis

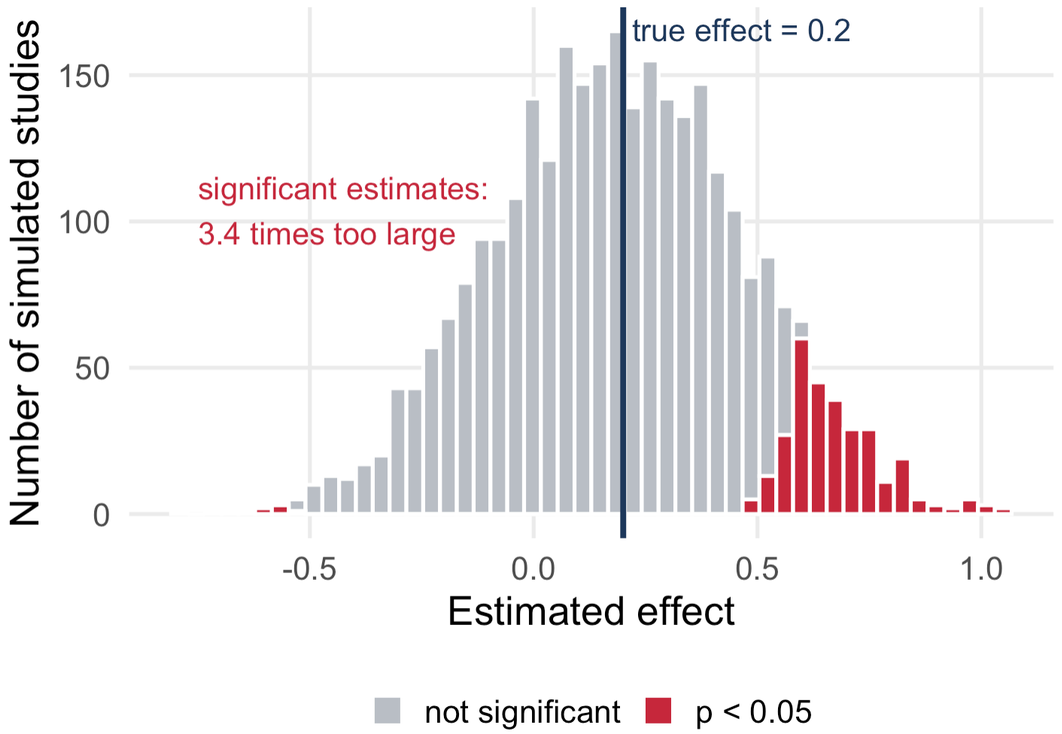


3,000 simulated survey experiments, 50 respondents per arm, true effect 0.2 SD, outcome measured with reliability about 0.5. Red: the studies that reach p < 0.05, and would get published.

Noisy measures and the replication crisis

Gelman & Carlin (2014): a statistically significant estimate can still be wrong in two ways.

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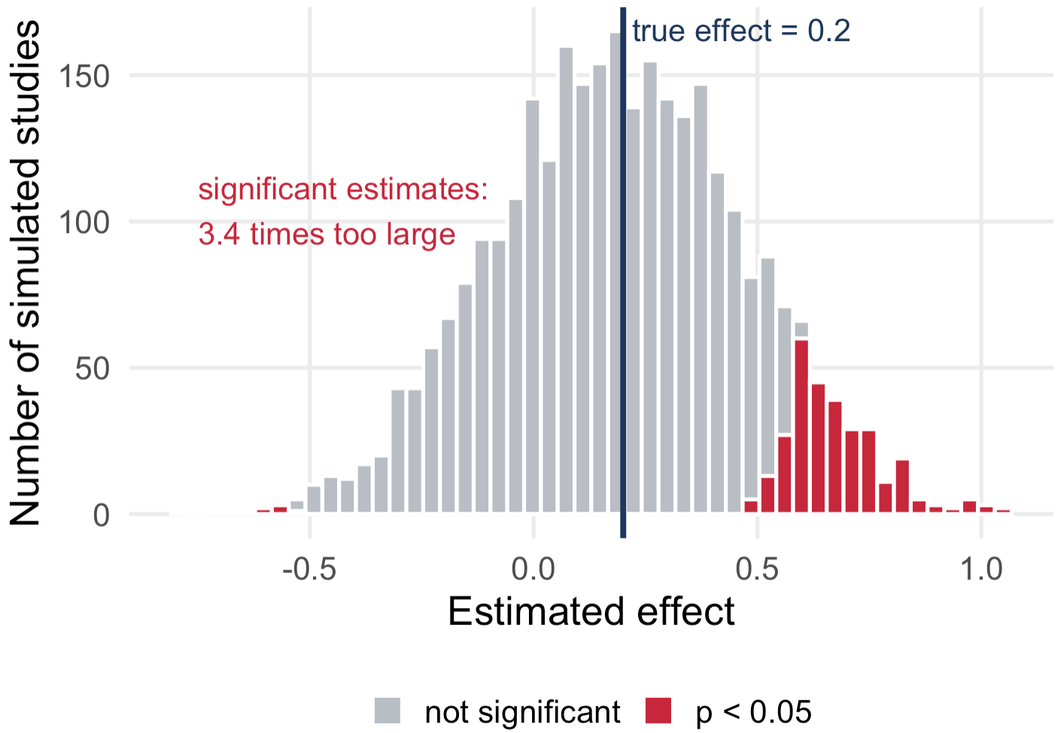
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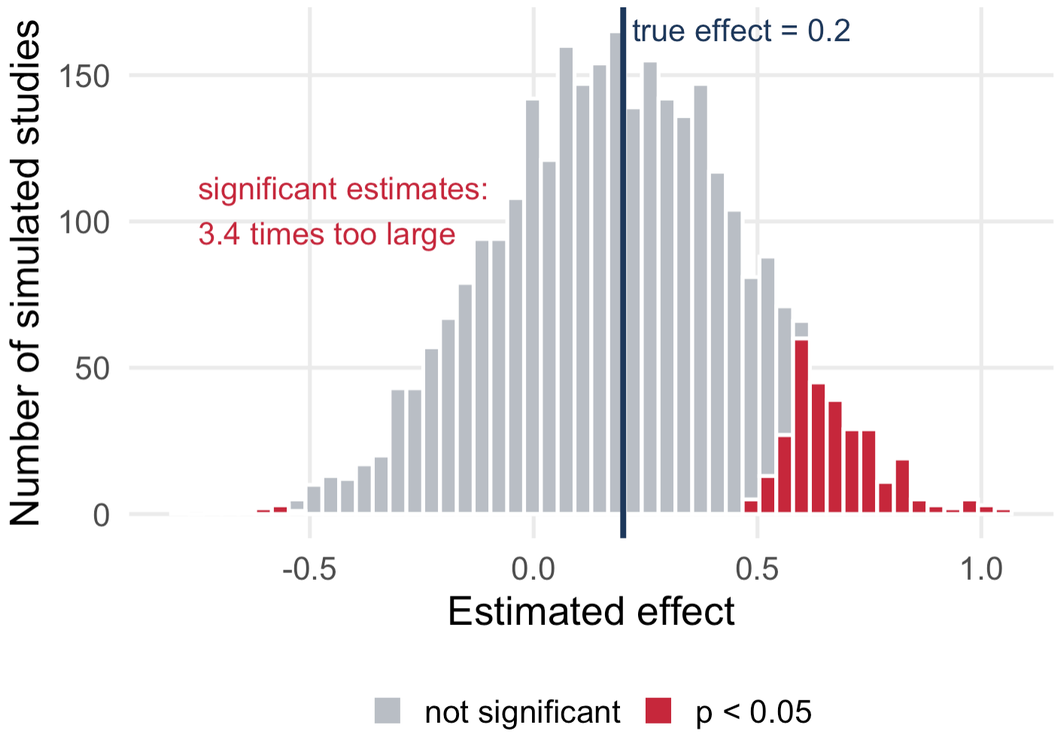
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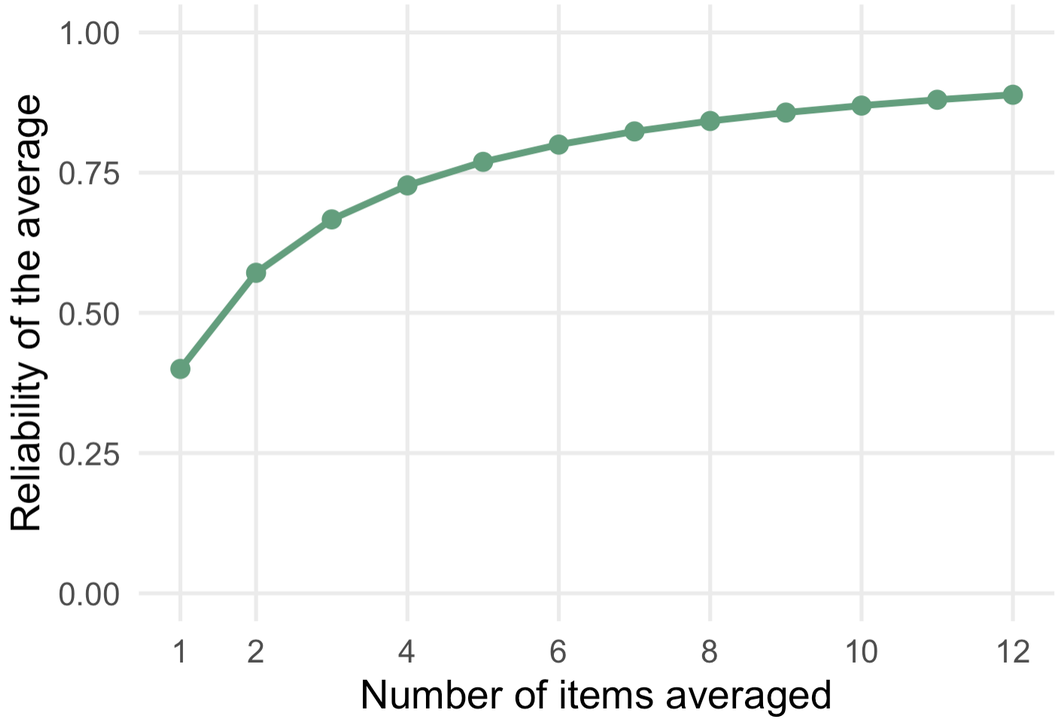
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- Common intuition: "noise makes effects *harder* to find, so a significant result despite noise must be strong."
- Wrong in small, noisy studies: only estimates that happen to be large pass $p < 0.05$, so the published ones **exaggerate the effect** (Loken & Gelman 2017).



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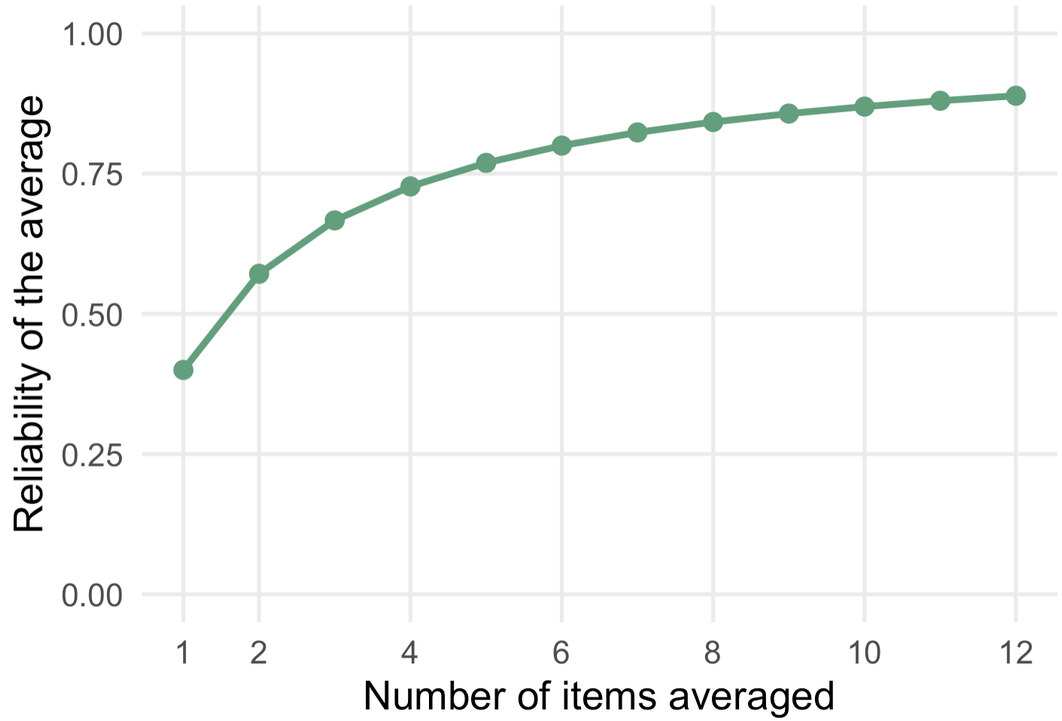
More indicators, less noise



Reliability of an average of k items, each with reliability 0.4 (Spearman–Brown formula). Most of the gain comes from the first few items.

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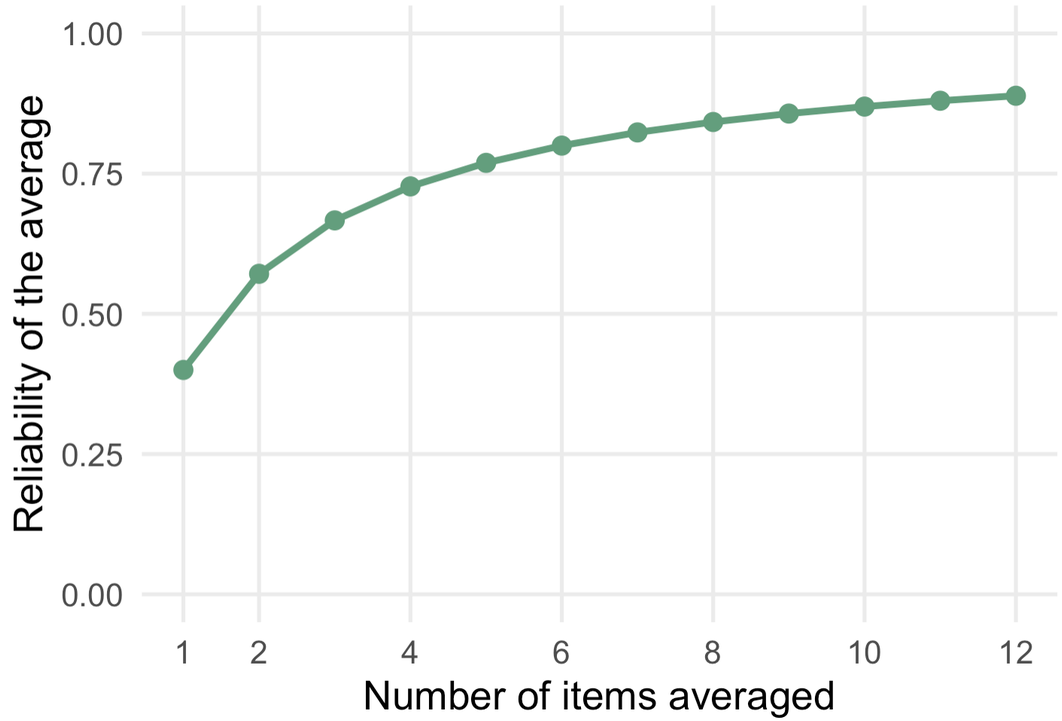
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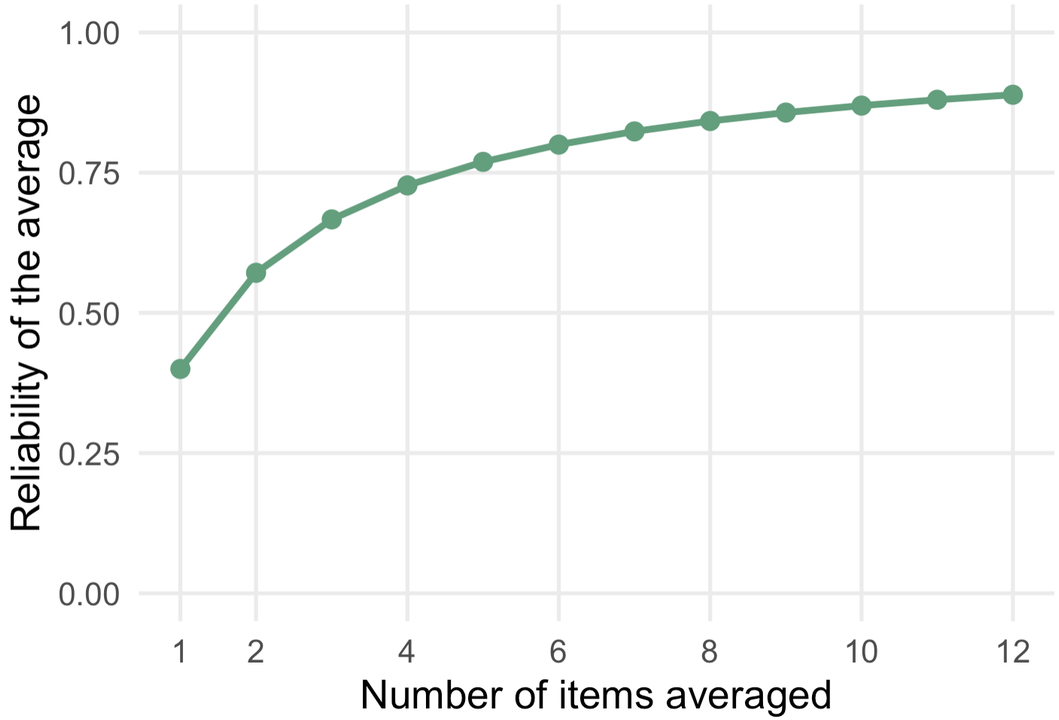
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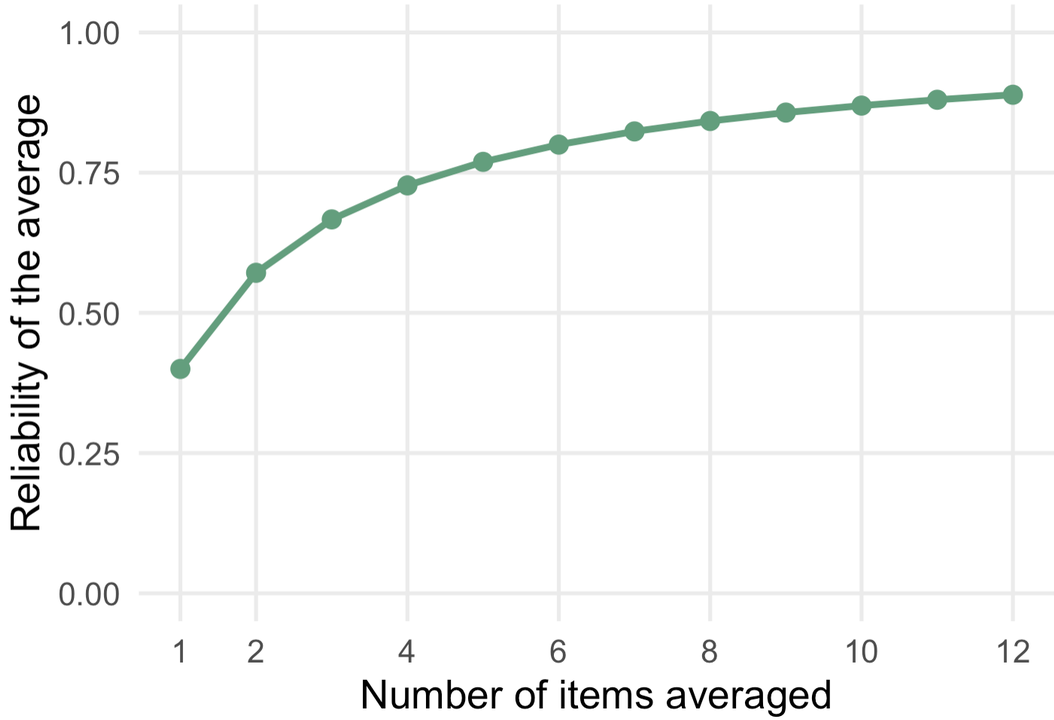
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- Next week: do this with a **model** of the construct.



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Your turn 🔍 (~5 min, solo, then show of hands)

For each study: is the estimated effect biased **toward zero**, **away from zero**, **not biased**, or **unclear**? Draw the DAG if it helps.

1. Effect of economic hardship on turnout. Hardship is one survey item that respondents answer somewhat at random.
2. Effect of austerity on protest. Protest events are hand-coded from newspapers, and coders make random mistakes.
3. Effect of a degree on immigration attitudes, controlling for family background with “books at home”.
4. A get-out-the-vote experiment that measures turnout by asking respondents whether they voted.

Does it travel?

“All generalizations are dangerous, even this one.”

attributed to Alexandre Dumas

Five dimensions of external validity

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X

Units

Would other people or places respond the same way?

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X

Units

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T

Treatments

Is the treatment elsewhere the same thing?

Five dimensions of external validity

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Units

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T

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Y

Outcomes

Is the outcome measured the same way, and does it mean the same thing?

Five dimensions of external validity

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- Most studies discuss external validity in a sentence. These dimensions turn it into **questions you can answer with data**.

Don’t be fooled by “representative”

IFLSCIENCE!

Survey Finds Most Americans Think That They Have Above Average Intelligence

DON'T SCOFF: YOU'RE NOT INVULNERABLE TO SELF-DELUSION. PR IMAGE FACTORY/SHUTTERSTOCK

A new US-based nationally representative survey has found that 65 percent of respondents (70 percent in men, 60 percent in women) agree with this rather telling statement: “I am more intelligent than the average person.” Hopefully this doesn’t require a rudimentary lesson in statistics to explain why this simply isn’t possible.

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Instead, it’s important because similar research conducted in the US half a century earlier found much the same thing. Although the researchers caution about generalizing their findings, it’s a good bet the same pattern can be found in other countries around the world too.



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31 JUL 2018, 14:55

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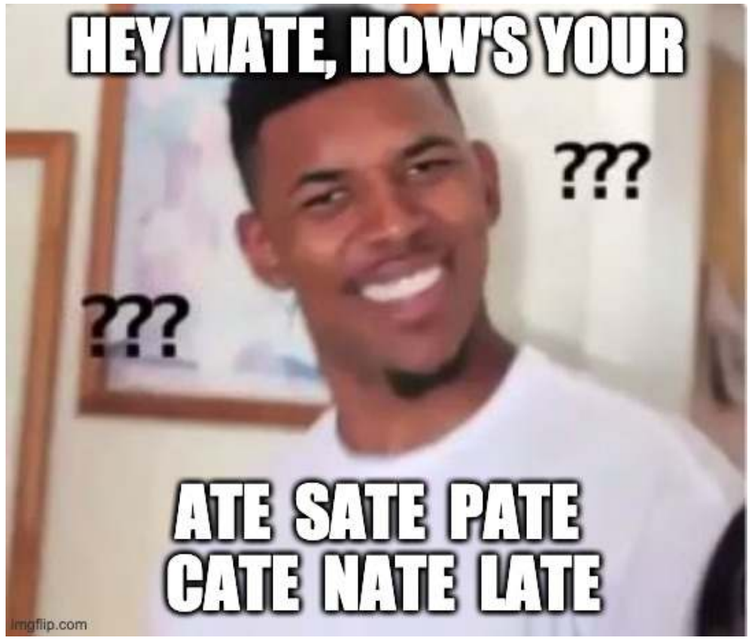


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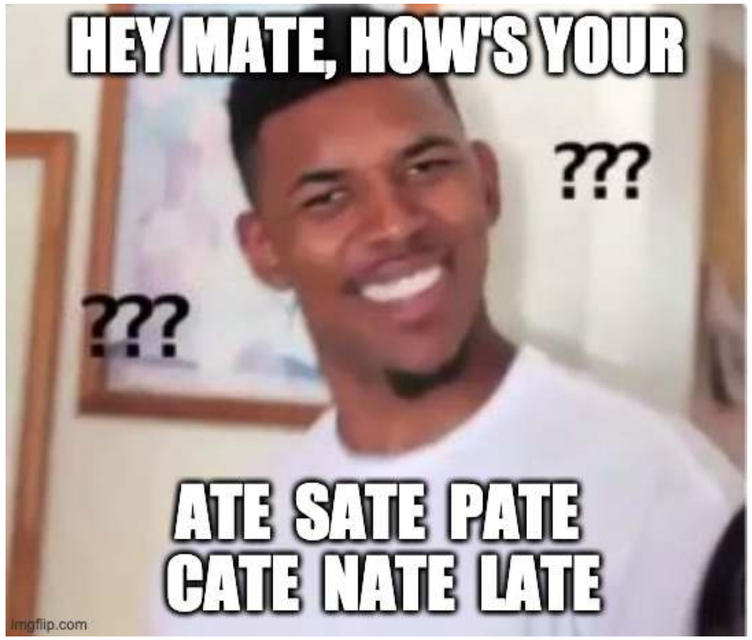
> Example: millions of rides logged in a fitness-tracking app (e.g., Strava) look like a huge sample of cycling behavior. They are a huge sample of *app users*: younger, sportier, more often male than the average commuter.

SATE vs PATE



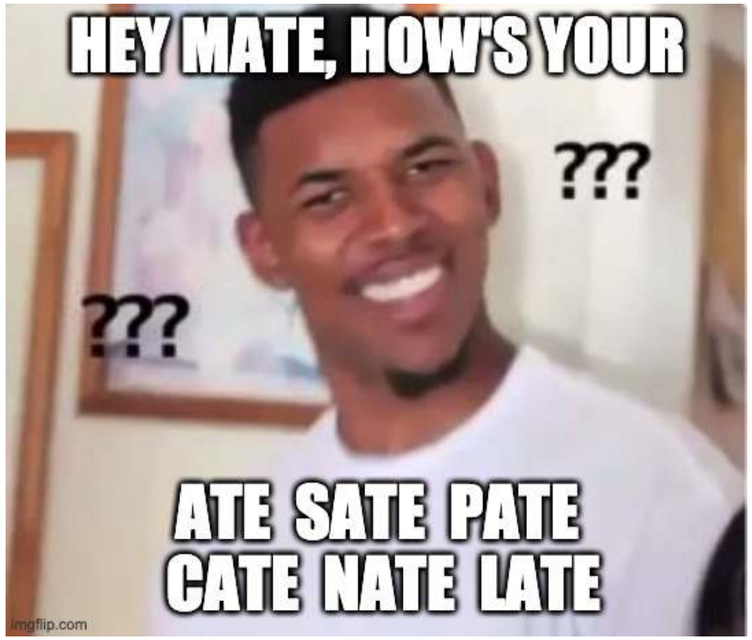
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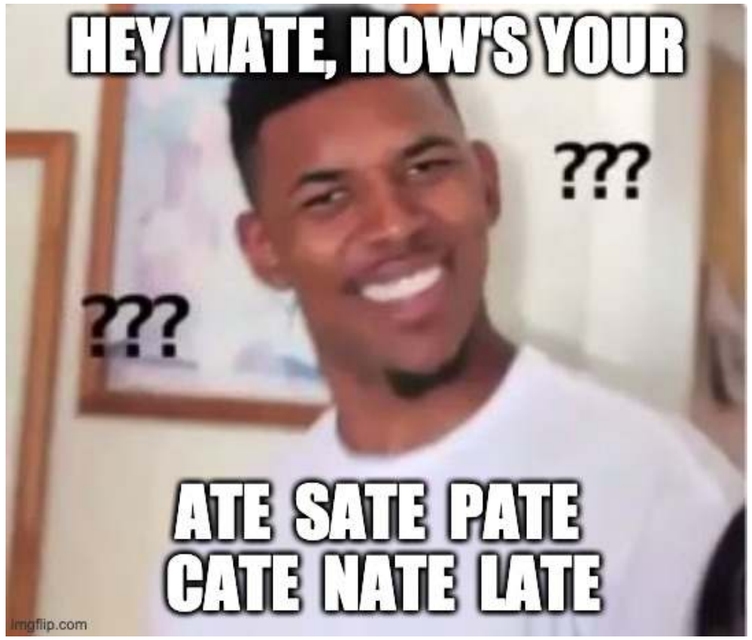
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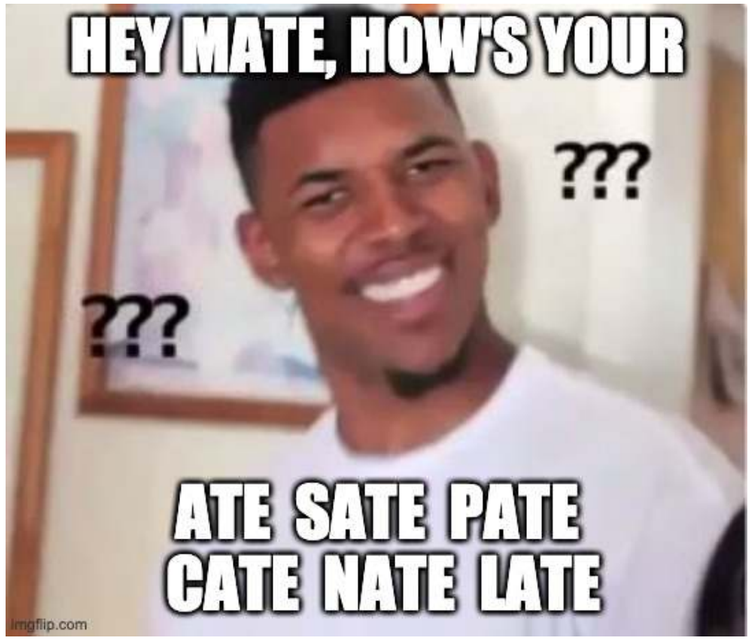
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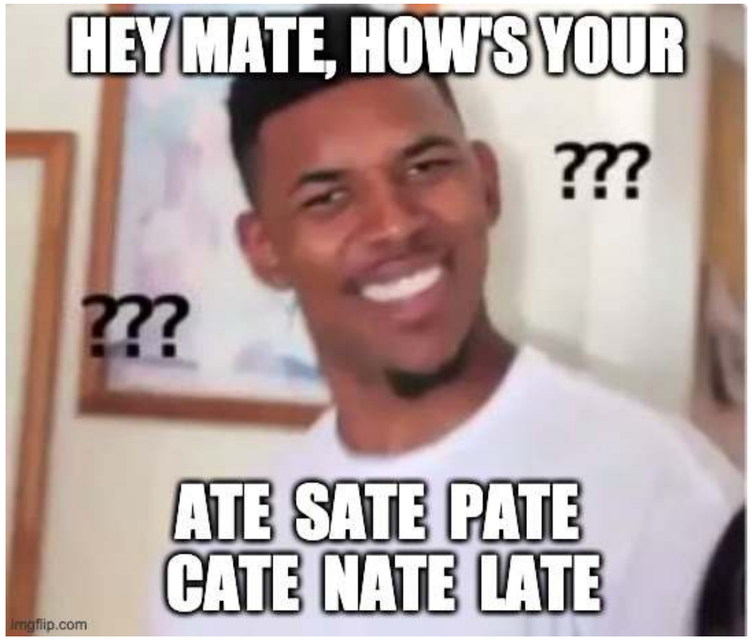
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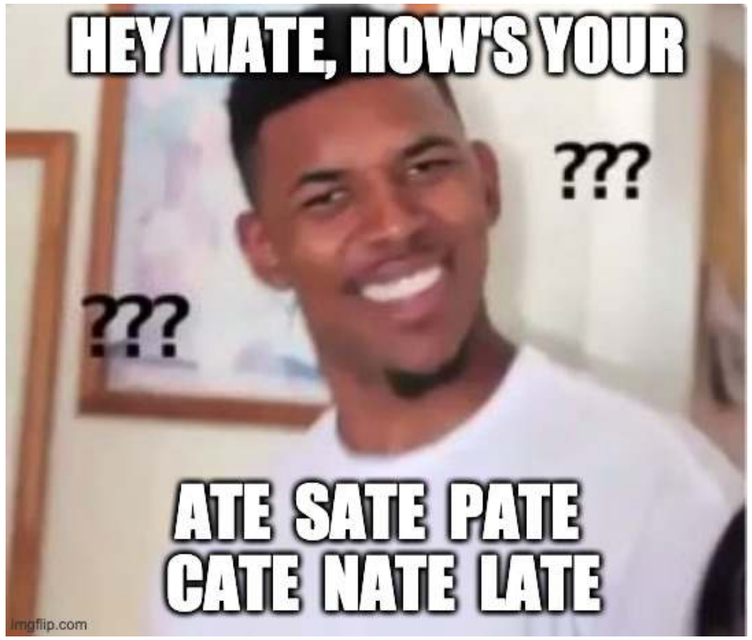
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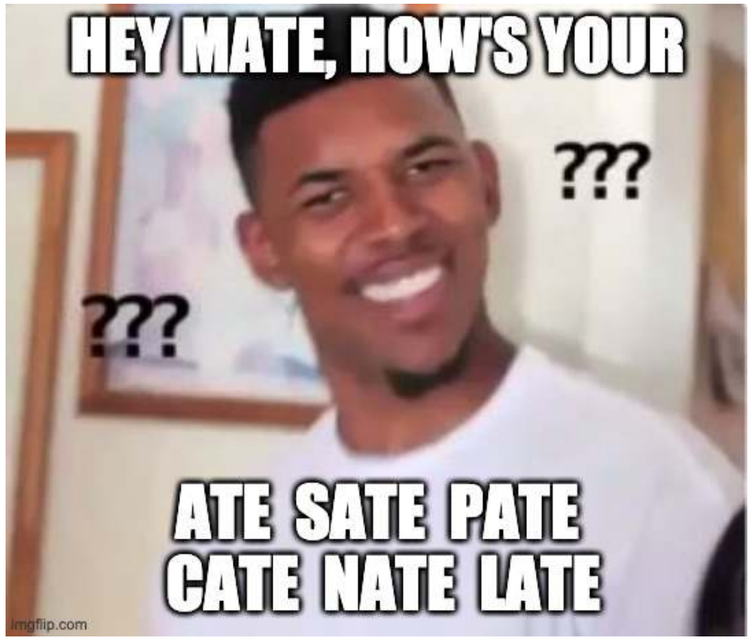
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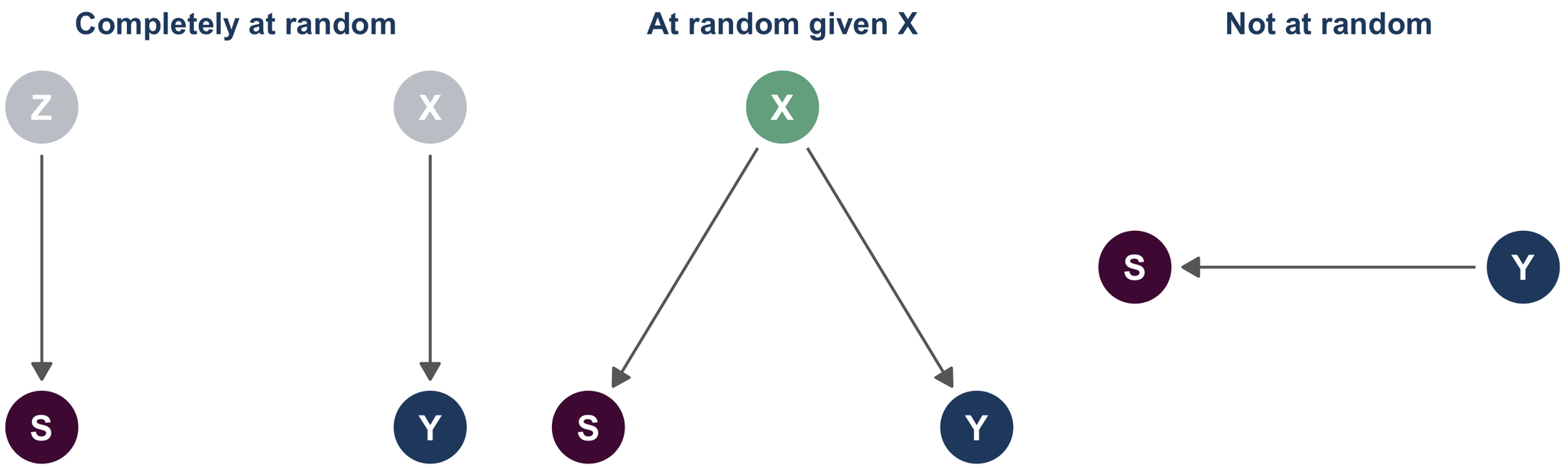


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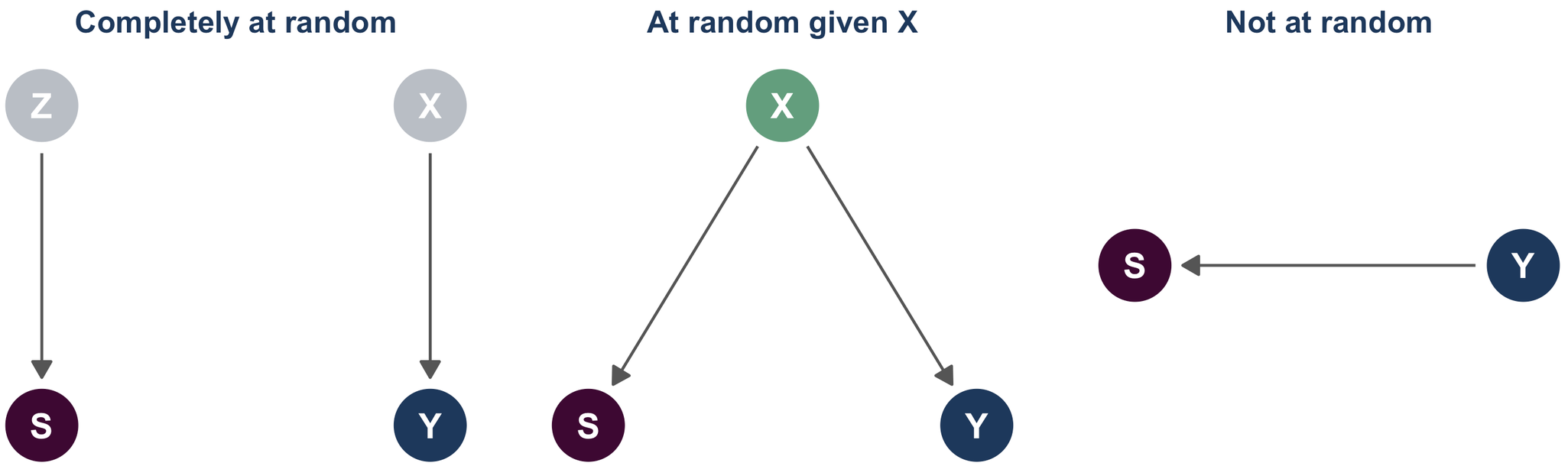
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- Week 3 obsessed over getting the ATE right *in the study*. Today: whose ATE?



Selection into the study is a DAG problem

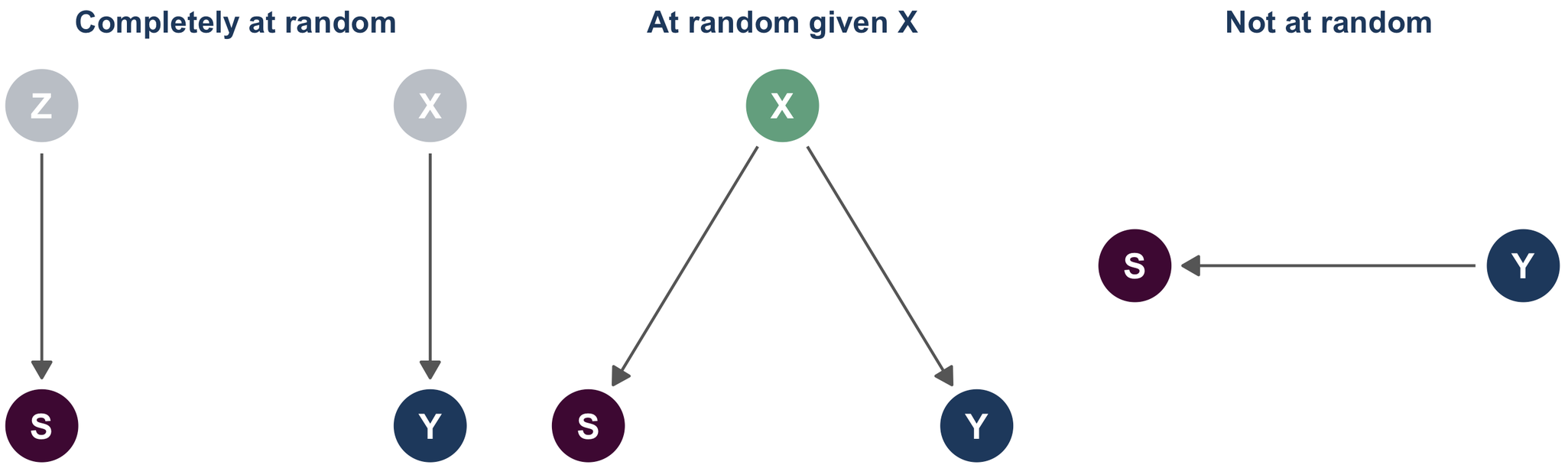


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- Same logic as Week 3's colliders: selecting the sample is conditioning on S.

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Effect of a protected lane by commute distance

<3 km: +10 · 3–8 km: +6 · >8 km: +1 pp

Copenhagen (study city): 50 / 40 / 10%

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- Requires: the modifiers are known, measured in **both** places, and present in both (overlap) (Pearl & Bareinboim 2014).

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A well-identified effect is an answer for **one population, one treatment, one outcome, one context, one time**.

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- Treat time as an effect modifier: **when** was the finding estimated, and has anything that moderates it changed since?

When the measure drifts: is the world less repressive?

Fariss (2014), "Respect for human rights has improved over time", *APSR* 108(2): 297–318

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READING 1

Repression did not decline

The scores are right; the world is stuck.

READING 2

The yardstick got stricter

The same practices get worse scores later.

Fariss (2014), "Respect for human rights has improved over time", *APSR* 108(2): 297–318

A changing standard of accountability

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Look harder

More information per country: more NGOs, more sources, longer reports.

A changing standard of accountability



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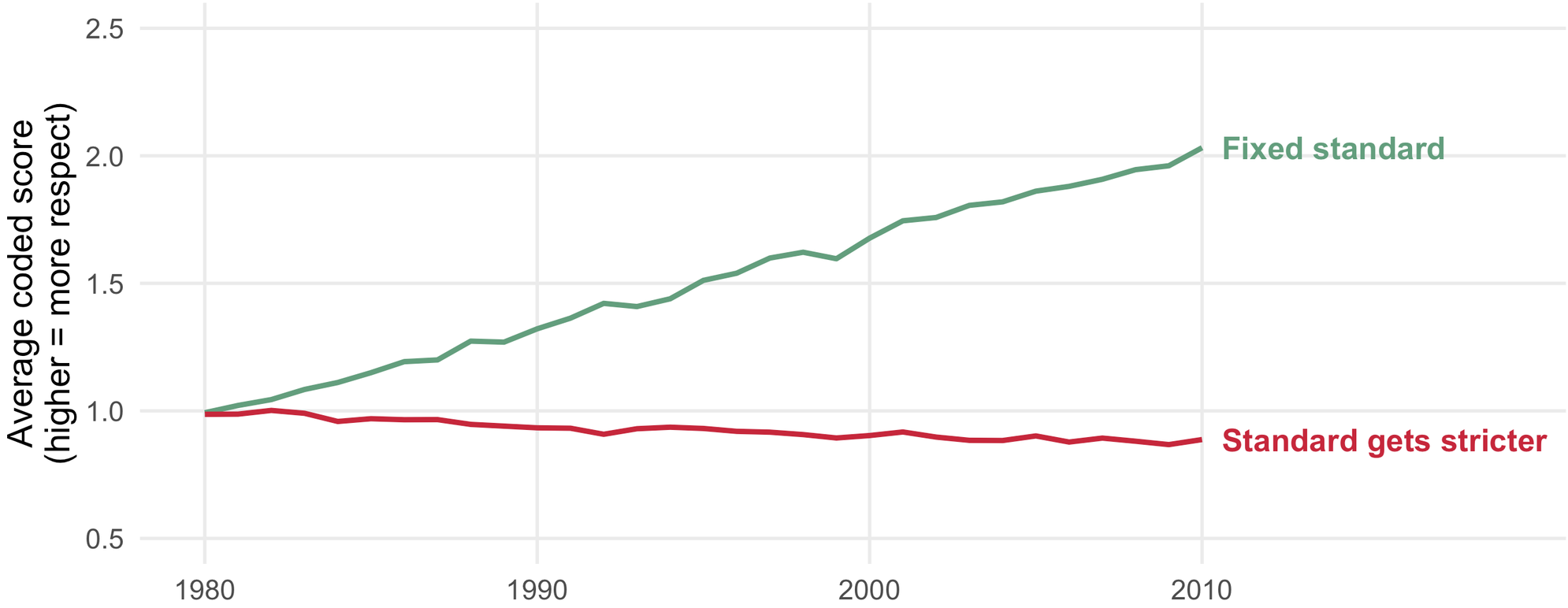


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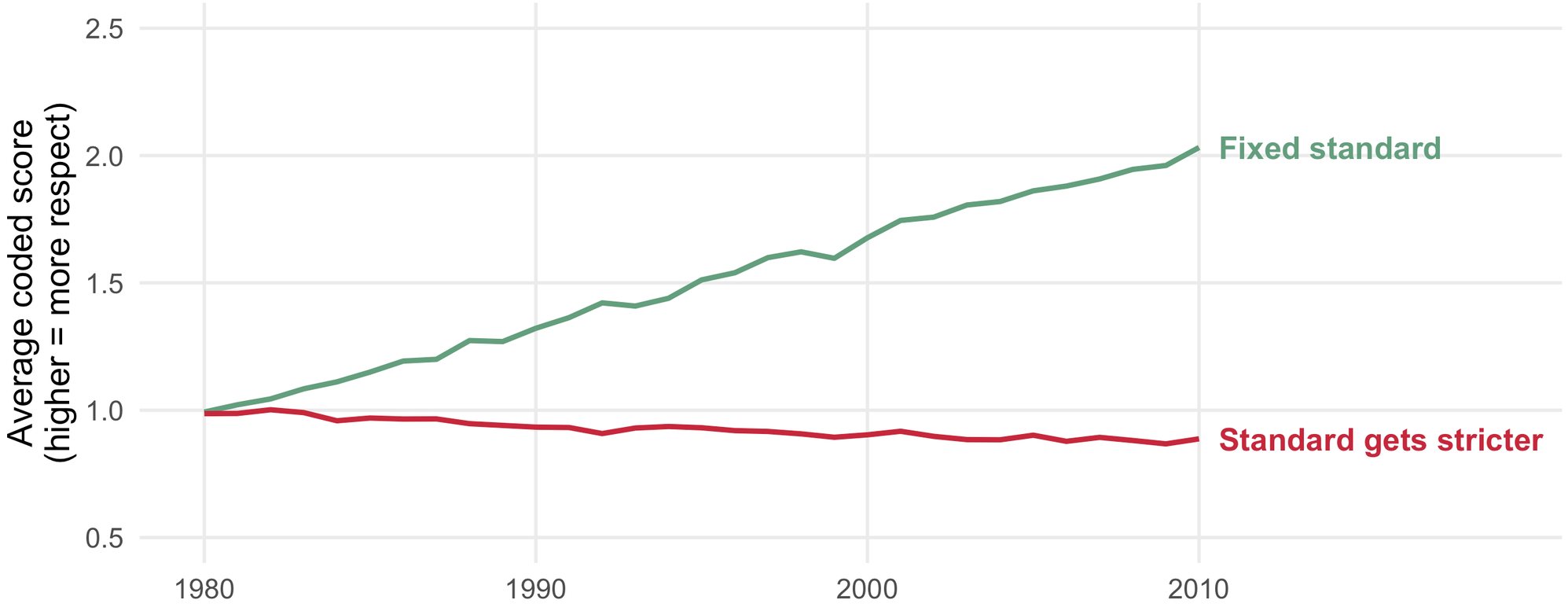
Same act, worse score: **the meaning of a “2” changed** between 1980 and 2010.

What a stricter yardstick does to the scores



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What a stricter yardstick does to the scores



In both lines, true respect for rights **improves at the same rate**. Only the 'yardstick' differs.

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Next week: does a measure **mean the same thing** across groups, countries, and years? That is **measurement invariance**: transportability for measures.

Your turn 🔍 (~6 min, groups of 3)

Each group takes one finding and a new target. Which dimension (**X, T, Y, C, time**) is the biggest threat? What data would you need to transport the effect?

1. Mailers showing your neighbors' voting records raise turnout in a 2006 Michigan primary → a 2026 Italian general election
2. Ten-minute "deep canvassing" conversations reduce transphobia in Miami, 2015 → attitudes toward refugees in Dresden, 2026
3. Anonymous CVs in voluntary French firms → mandatory anonymization in Italy
4. Vouchers to move to low-poverty neighborhoods raise children's adult earnings in US cities → social housing in Paris or Milan

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- The core problem is **transparency**: if measurement decisions are not reported, nobody can evaluate them, including you.

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- Most QMPs are not fraud. They are undisclosed **forking paths** (Week 1), and they make findings impossible to evaluate or compare.

How flexible can a measure be?

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150+

ways to score one aggression task

The "competitive reaction time task" has been quantified in more than 150 different ways across published studies (Elson et al. 2014). Pick the scoring after the fact, and almost any result is reachable.

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Seven common depression scales cover 52 different symptoms, with little overlap (Fried 2017). "Depression" in two papers can be two different constructs.

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📖 The task comes from research on violent video games. A good opportunity to plug Kevin’s blog… In this case about what a literature can and cannot ask: Does pinball cause violence? and Video games cause Elon Musk.

Six questions to answer about every measure

Flake & Fried's recommendation is simple: **report the answers**.

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> 1 What is your construct?

Six questions to answer about every measure

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1. What is your construct?
2. Why and how did you select your measure?

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1 What is your construct?

2 Why and how did you select your measure?

3 What measure did you use to operationalize the construct?

Six questions to answer about every measure

Flake & Fried's recommendation is simple: **report the answers**.

1. What is your construct?
2. Why and how did you select your measure?
3. What measure did you use to operationalize the construct?
4. How did you quantify your measure?

Six questions to answer about every measure

Flake & Fried's recommendation is simple: **report the answers**.

1. What is your construct?
2. Why and how did you select your measure?
3. What measure did you use to operationalize the construct?
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5. Did you modify the scale? How and why?

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Measurement is a neglected part of the replication crisis: a well-identified effect on a poorly understood measure is not a finding about the concept.

Five questions before you trust a number

A pre-flight checklist for any measure in a paper you read or write:

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- **What is the systematized concept?** Is it stated, and does it match the claim in the abstract?

Five questions before you trust a number

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- **For whom and when was it validated?** Does that match your population and period?
- **Was it modified?** Items dropped, recoded, or combined, and is that reported?

What we covered today

FROM CONCEPTS TO NUMBERS, AND BEYOND THE STUDY

Concepts to indicators

Background concept → systematized concept → indicators → scores; critiques target a specific link

Validity is an argument

Validation targets the systematized concept; five sources of evidence for an interpretation, a population, and a purpose

Measurement error

Error in the treatment attenuates; in a control, leaves residual confounding; when differential, biases in any direction

Does it travel?

External validity along X, T, Y, C, and time; transport by reweighting effect modifiers; measures and findings can expire; report every measurement decision (Flake & Fried)

Readings

Compulsory

- Adcock & Collier (2001). Measurement validity: A shared standard for qualitative and quantitative research. *APSR* 95(3): 529–546.
- Flake & Fried (2020). Measurement schmeasurement: Questionable measurement practices and how to avoid them. *Advances in Methods and Practices in Psychological Science* 3(4): 456–465.
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Strongly encouraged

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- Munger (2023). Temporal validity as meta-science. *Research & Politics* 10(3).

Optional

- Findley, Kikuta & Denly (2021). External validity. *Annual Review of Political Science* 24: 365–393.
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- Wooldridge, Ch. 9 (errors-in-variables).

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- Vavreck (2007). *Quarterly Journal of Political Science* 2(4): 325–343.

Every number in your dataset is a claim.

Your job is to know what it claims to measure,

and where that claim stops being true.

Next week: Measurement II · latent variables

In the lab on Monday

with Martín

In Lab 4 you'll build a trust-in-institutions scale and compute Cronbach's α by hand, estimate test–retest and inter-coder reliability, simulate attenuation, residual confounding, and misreported turnout, and reweight a bike-lane effect to a new city.

`across()` `rowMeans()` `purrr::map()` `cut()` `weighted.mean()`

LAB 4

`04-measurement-lab.qmd`